

ESSENTIAL MATHEMATICS GRADE 10: REFLECTION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 2.2 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 2.0: Measurements and Geometry
Sub-Strand	Sub-Strand 2.2: Reflection
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Symmetry and lines of symmetry in plane figures – Properties of reflection using objects and their images – Reflection on a plane surface (drawing images given an object and a mirror line) – Reflection on a Cartesian plane (mirror lines $x=0$, $y=0$, and $y=x$) – Drawing the mirror line given an object and its image (plane surface and Cartesian plane) – Application of reflection in real-life situations (e.g., driving mirrors, road safety, design patterns)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - identify lines of symmetry in plane figures, - determine the properties of reflection using objects and its image, - draw an image given an object and a mirror line on a plane surface, - draw an image given an object and a mirror line on a Cartesian plane, - draw the mirror line given an object and its image on a plane surface and a Cartesian plane, - apply reflection in real-life situations, - appreciate the role of reflection in real-life situations.
Core Competencies	– Learning to Learn: The learner develops relationship skills when discussing with peers the applications of reflections in real life such as use of mirrors in road transport. – Creativity and Imagination: The learner acquires connection skills when making cut outs of plane figures for finding the number of lines of symmetry.
Core Values	– Unity: The learner takes turns when working with peers to make cut outs of plane figures used in reflection activities. – Responsibility: The learner takes care of instruments used to construct images formed after reflection in the mirror lines.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners question why an image in a mirror appears the same distance behind the mirror as the object is in front, driving inquiry into the properties of reflection. – Planning and Carrying Out Investigations: Learners systematically place objects at measured distances from a plane mirror and record image positions to deduce reflection properties. – Analyzing and Interpreting Data: Learners measure and compare object and image distances, perpendicularity of the mirror line, and size congruence to draw evidence-based conclusions.

	– Constructing Explanations: Learners use gathered evidence from plane and Cartesian-plane activities to articulate the invariant properties of reflection (congruence, perpendicular bisector relationship). – Obtaining, Evaluating, and Communicating Information: Learners discuss with peers and present findings on real-life applications of reflection such as driving mirrors and road safety.
Pertinent & Contemporary Issues (PCIs)	– Road Safety: The learner discusses with peers the applications of reflections in real life such as use of driving mirrors in reducing road crashes on Kenyan roads.
Career Connections	– Architecture and Interior Design: Reflection symmetry is used in designing buildings, floor tile patterns, and décor — relevant to construction careers prevalent across Nairobi and other urban centres. – Road Transport and Logistics: Matatu and truck drivers rely on plane and convex mirrors; understanding reflection supports careers in vehicle operation and road safety management. – Fashion and Textile Design: Kanga, kitenge, and Maasai shuka pattern designers use lines of symmetry and reflection to create repeating motifs. – Photography and Media: Photographers and videographers use mirror reflections for composition; media technology students benefit from understanding image formation. – Sports Coaching and Recreation: Coaches use reflective analysis of athletes' movements; sports management careers at institutions such as those supporting Kenyan marathon runners benefit from spatial reasoning. – Guidance and Counselling / Health Sciences: Body-symmetry assessments (e.g., physiotherapy) and use of diagnostic mirrors draw on knowledge of reflection.
Focus for Lessons	This unit develops learners' understanding of reflection as a geometric transformation, beginning with lines of symmetry in familiar Kenyan objects and progressing to formal constructions on plain paper and the Cartesian plane. Learners move from hands-on exploration with physical mirrors and cut-out shapes to precise geometric construction, always connecting abstract properties back to the anchoring real-world context of mirror images encountered in daily Kenyan life. Throughout the eight lessons, collaborative investigation, peer discussion, and step-by-step model building ensure that both procedural fluency and conceptual understanding of reflection are achieved.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: Why does a mirror show a perfect copy of the world — and where exactly does that copy live? KICD KEY INQUIRY QUESTION: 1. How is reflection important in day-to-day activities?
Anchoring Phenomenon	A Grade 10 learner in Nairobi notices something puzzling one morning: as she stands in front of the full-length mirror on her bedroom wall to check her school uniform, her reflection raises the opposite hand when she raises hers, yet every detail — the pattern on her Kanga skirt, the number of buttons on her shirt, the distance she stands from the glass — appears perfectly preserved. Later, on the matatu ride to school, she observes the driver constantly checking the rear-view and side mirrors; the vehicles and pedestrians visible in those mirrors are clear, life-sized representations of what is behind, yet they appear "flipped." At school, the mathematics teacher holds up a sheet of paper with a bold letter "A" printed on it and places a plane mirror beside it: the class sees a perfect mirror-image "A" in the glass — same size, same shape, but laterally reversed. Students are puzzled: Why does the image appear the same distance behind the mirror as the object is in front? Why is the image the same size but "flipped"? What invisible line or rule governs where the image lands? This observable, puzzling, everyday phenomenon — the behaviour of plane mirrors — anchors the entire unit and motivates learners to uncover the precise geometric properties of reflection.
Storyline Thread	Lesson 1 — Symmetry Safari (Lines of Symmetry in Plane Figures): Learners are introduced to the anchoring phenomenon (the classroom plane mirror and the "flipped letter A"). They collect leaves, cut-out shapes (rectangles, squares, triangles, circles), and everyday objects (Kanga fabric swatches, bottle caps) and investigate lines of symmetry by folding and using a plane mirror. They

record the number of lines of symmetry for each shape and share findings with peers, generating initial questions for the Driving Question Board.

Lesson 2 — Mirror, Mirror on the Wall (Properties of Reflection — Plane Surface): Learners stand at measured distances from a plane mirror and record the apparent distance of their image. They then use a ruler and set square to place a small object at various distances from a drawn mirror line on plain paper, locate the image, and measure: (i) object distance = image distance from mirror line; (ii) the line joining object to image is perpendicular to the mirror line; (iii) the image is congruent to the object. Learners formalise the three properties of reflection and connect them back to the anchoring phenomenon.

Lesson 3 — Drawing Images on a Plane Surface (Construction Practice): Learners use a geometrical set to construct images of given plane figures (triangles, quadrilaterals, irregular polygons) reflected in horizontal, vertical, and oblique mirror lines on plain paper. Step-by-step teacher modelling is followed by peer practice. Learners annotate each construction to label object distance, image distance, and perpendicular bisectors, reinforcing properties from Lesson 2.

Lesson 4 — Reflection on the Cartesian Plane I ($x=0$ and $y=0$): Learners plot given objects on a Cartesian plane and reflect them in the y -axis ($x=0$) and x -axis ($y=0$). They discover the coordinate rules: reflection in $x=0$ maps $(x, y) \rightarrow (-x, y)$; reflection in $y=0$ maps $(x, y) \rightarrow (x, -y)$. Learners connect these rules to the mirror-image phenomenon — the mirror line is now a coordinate axis — and practise with triangles and quadrilaterals plotted using Kenyan context coordinates (e.g., grid maps of school grounds in Nairobi).

Lesson 5 — Reflection on the Cartesian Plane II ($y=x$): Learners investigate reflection in the line $y=x$ by plotting objects and their images and deducing the rule $(x, y) \rightarrow (y, x)$. They compare all three Cartesian mirror lines studied ($x=0$, $y=0$, $y=x$), tabulate the coordinate transformation rules, and practise mixed examples. The lesson revisits the anchoring phenomenon: just as standing closer to a physical mirror changes the image position, changing the mirror line on the Cartesian plane changes the image coordinates by a precise rule.

Lesson 6 — Finding the Mirror Line (Reverse Reflection): Given an object and its image on both a plain surface and a Cartesian plane, learners work backwards to construct or identify the mirror line (perpendicular bisector of the segment joining corresponding points). This reverse-engineering task deepens understanding of the properties of reflection. Learners connect this to the road-safety supporting phenomenon: traffic engineers determine the optimal angle and position of a junction mirror by knowing where drivers (object) and the hazard (image) must be.

Lesson 7 — Real-Life Applications and Peer Presentations: Learners investigate real-life applications of reflection in groups: (i) driving mirrors and road safety on Kenyan roads; (ii) symmetry in Kanga/kitenge design; (iii) architectural symmetry in Kenyan buildings (e.g., Parliament buildings, mosques in Mombasa). Each group prepares a short illustrated presentation linking their application to the geometric properties of reflection studied in Lessons 2–6. The Driving Question Board is updated with answers and residual questions.

Lesson 8 — Model Building, Review and Assessment: Learners produce a cumulative annotated diagram ("reflection model") that synthesises all properties and constructions: lines of symmetry, plane-surface reflection, Cartesian reflection in $x=0$ / $y=0$ / $y=x$, and reverse mirror-line construction. They self-assess against the KICD rubric descriptors, complete a class activity (written exercise covering all SLOs), and reflect on how the anchoring phenomenon of the everyday mirror is now fully explained by the geometric properties they have uncovered across the unit.

Supporting Phenomena	– Symmetry in Kenyan leaves and national symbols: Learners collect leaves (e.g., mango, avocado, neem) from the school compound in Kisumu or Nairobi and notice that folding many leaves along their central vein produces two identical halves — raising the question of what geometric rule produces this symmetry. – Reflections on Lake Victoria and other water bodies: Photographs taken at Lake Victoria's shore or the Nairobi Dam show perfect upside-down reflections of trees, boats, and buildings on calm water — puzzling learners about why the reflected image appears directly below the object. – Kanga and kitenge textile patterns: Learners examine the repeating symmetric patterns on Kanga cloth sold in Mombasa markets and notice that half the pattern is a mirror image of the other half, raising questions about how designers use lines of reflection to create such patterns. – Road safety mirrors at Kenyan junctions: Convex mirrors mounted at sharp bends on Kenyan rural roads (e.g., along the Kericho–Kisumu highway) show wide-angle reflections of oncoming traffic, prompting discussion about how the position and angle of a mirror line determines what is visible in the image.
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ATTRIBUTION AND LICENSING

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Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (40 minutes): Symmetry Safari: Lines of Symmetry in Plane Figures

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To introduce the anchoring phenomenon of reflection through hands-on investigation of lines of symmetry in plane figures, and to launch the Driving Question Board.
Knowledge	Learners identify lines of symmetry in plane figures including rectangles, squares, equilateral triangles, circles, and irregular shapes such as Kanga fabric cut-outs and leaf specimens.
Skills	Learners make cut-outs of plane figures, fold them to locate lines of symmetry, use a plane mirror to verify symmetry, record findings in a table, and communicate observations to peers.
Attitudes	Learners appreciate the presence of symmetry in everyday Kenyan objects and environments, and develop curiosity about why mirror images appear 'flipped' yet perfectly preserved.
Key Inquiry Question	How is reflection important in day-to-day activities?
Purpose in Storyline	This is the opening lesson of the unit. It anchors the entire storyline by presenting the puzzling phenomenon of the plane mirror and the 'flipped letter A' observed in the classroom. Learners conduct a Symmetry Safari — investigating lines of symmetry in leaves, cut-out shapes, and Kanga fabric swatches — to build the foundational concept of a line of symmetry as an invisible geometric rule that governs where a mirror image lands. Their questions and observations seed the Driving Question Board that will grow across all lessons in this unit.
Safety Notes	Learners should handle scissors carefully when making cut-outs. Plane mirrors (if glass) should be handled with care to avoid breakage; acrylic mirrors are preferred. No sharp tools should be left unattended on desks.

B. LESSON OVERVIEW

In this lesson, learners are introduced to the anchoring phenomenon: a Grade 10 learner in Nairobi notices that her reflection in a full-length mirror raises the opposite hand, and her mathematics teacher places a bold letter 'A' next to a plane mirror to reveal a laterally reversed image. Learners are immediately puzzled — why does the mirror produce a copy that is 'flipped'? What invisible line or rule governs this? To begin answering these questions, learners conduct a Symmetry Safari: they collect leaves from the school compound, use pre-cut paper shapes (rectangles, squares, equilateral triangles, circles), and examine Kanga fabric swatches and bottle caps. They fold and use a plane mirror to locate and count lines of symmetry, recording results in a class table. The lesson closes with learners generating initial questions for the Driving Question Board, which will be revisited and expanded across the entire unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reflection Symmetry	Learners observe the teacher hold up a sheet of paper with a bold letter 'A' printed on it and place a plane mirror beside it. Before	1. Place the letter 'A' sheet and plane mirror at the front. Say: 'Before I show you anything — look carefully and write your	Think-Pair-Share with written predictions. This strategy externalises prior knowledge and creates cognitive dissonance —	Teacher circulates during individual writing and notes: (a) whether learners draw the mirror-A correctly reversed or not, (b)

Principles - Basic Source: CK-12 Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12 Search ARES for similar readings	being told anything, learners individually write predictions in their exercise books responding to three prompts: (1) 'Draw what you think the mirror image of the letter A will look like.' (2) 'If I stand 30 cm from the mirror, how far behind the mirror do you think my image will appear? Why?' (3) 'Look at the Kanga swatch on your desk — do you think it has any lines of symmetry? How many?' Learners then share predictions in pairs (Think-Pair-Share) for 2 minutes before two or three pairs are cold-called to share with the class.	prediction silently. You have 90 seconds.' [WAIT 90 seconds — circulate, do NOT give hints.] 2. Display the three written prompts on the board. Read each aloud. 3. After individual writing, say: 'Now turn to your partner. Share your drawings and predictions. You have 2 minutes.' [WAIT 2 minutes.] 4. Cold-call 3 pairs: 'Amina and Kevin — what did you predict for prompt two and why?' — 'Wanjiru and Otieno — how many lines of symmetry did you predict for the Kanga swatch?' — 'Akinyi and Mwangi — what does your drawing of the mirror-A look like?' 5. Do NOT confirm or deny predictions. Say: 'Hold those predictions. We are going to gather evidence right now and see who was closest.'	when learners see the actual mirror image, the mismatch between prediction and observation drives curiosity and motivates the investigation. Written predictions also give the teacher a quick formative snapshot of baseline understanding.	whether distance predictions show any proportional reasoning, (c) whether Kanga symmetry estimates are wild guesses or show some geometric intuition. Teacher mentally flags 2–3 learners with interesting misconceptions to cold-call during the Explain phase.
Observe Phase  VIDEO: Identifying symmetrical figures Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Identify line symmetry Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners work in groups of four for the Symmetry Safari. Each group receives: a set of pre-cut paper shapes (rectangle, square, equilateral triangle, circle, scalene triangle, regular hexagon), two leaves collected from the school compound, one Kanga fabric swatch (approx. 10 cm × 10 cm), one bottle cap, and one small acrylic plane mirror. Task 1 — Folding Test: Learners fold each shape and object to find all lines of symmetry, marking each fold line with a pencil. Task 2 — Mirror Test: Learners place the plane mirror along each proposed fold line to verify whether the reflected half matches the other half exactly. Task 3 — Recording: Groups complete a class results table (drawn in exercise books) with columns: Object/Shape Predicted lines of symmetry Lines found by folding Lines confirmed by mirror.	1. Distribute materials. Say: 'Your job is to be symmetry detectives. Use folding AND the mirror to find every line of symmetry in each object. Record everything in your table. You have 12 minutes — start now.' 2. At the 4-minute mark, circulate and prompt any group that is only folding and not using the mirror: 'Have you tried placing the mirror along that fold line? What do you notice when you do?' [WAIT 10 seconds for response before moving on.] 3. At the 8-minute mark, pose a challenge to groups that finish early: 'Can you find a shape with zero lines of symmetry? Can you find one with infinite lines of symmetry? Add those to your table.' 4. At the 12-minute mark, say: 'Pencils down. Each group, count the total number of different symmetry values you found across all shapes. We will share results now.'	Concrete-Representational-Abstract (CRA) via guided inquiry. Learners begin with physical folding (concrete), move to mirror verification (representational — the mirror creates an image), and record in a symbolic table (abstract notation). Connecting the plane mirror directly to the fold line builds the geometric intuition that a line of symmetry IS a mirror line — a key conceptual bridge to all subsequent lessons on reflection.	Observe group tables during circulation: (a) Does the group correctly identify that a circle has infinite lines of symmetry? (b) Does the group distinguish between a square (4 lines) and a rectangle (2 lines)? (c) Does the group correctly record 0 lines for the scalene triangle? (d) Is the mirror being placed on the fold line correctly, or is it being held at random angles? These observations reveal whether learners understand that a line of symmetry must produce a perfect match — the core property of reflection.

	Groups also sketch the letter 'A' mirror test result observed at the front of the class and add it to their table. Total activity time: 12 minutes.	5. Cold-call 4 groups to report one shape each. Ask: 'Group 3 — Fatuma, how many lines did the equilateral triangle have, and how did the mirror help you confirm that?' [WAIT 10 seconds.]		
Explain Phase  VIDEO: Identifying Lines of Symmetry in Letters Source: CK-12  Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12  Search ARES for similar readings	Groups share results and the class builds a consensus table on the board. The teacher then connects findings back to the phenomenon: the letter 'A' and the Kanga swatch are revisited. Learners are guided to articulate a working definition of a line of symmetry in their own words, then refine it using the mathematical language introduced by the teacher. Learners individually write a definition in their exercise books and answer two consolidation questions: (1) 'A regular hexagon has ___ lines of symmetry. Draw them.' (2) 'The letter A has ___ line(s) of symmetry. The letter S has ___ line(s) of symmetry. Explain the difference.' Learners then return to their original predictions and annotate them: 'I was right / I need to revise this because ___.'	1. Build the consensus table on the board by cold-calling one reporter per group. After each group reports, ask the class: 'Does anyone want to challenge or add to that?' [WAIT 10 seconds.] 2. Point to the letter 'A' on the board. Say: 'Based on your safari, Juma — where is the line of symmetry of letter A, and what does that tell us about the mirror image we saw?' [WAIT 15 seconds. If no response, prompt: 'Think about where you would fold the letter A so both halves match perfectly.'] 3. Introduce the formal definition: 'A line of symmetry divides a figure into two congruent halves that are mirror images of each other. We also call it a mirror line.' Write this verbatim on the board. Say: 'Copy this definition and then write it again in your own words.' 4. Cold-call 3 learners to read their own-words definitions aloud. Affirm correct use of 'congruent', 'halves', and 'mirror image'. 5. Connect to phenomenon: 'So when Aisha checked her uniform in the mirror this morning — which line of symmetry was the mirror acting as?' [WAIT 15 seconds. Cold-call 2 learners.] 6. Pose the consolidation questions. Give 4 minutes of silent individual work. Cold-call 2 learners to explain question 2.	Claim-Evidence-Reasoning (CER) discussion. Learners are asked to make a claim (e.g., 'A rectangle has 2 lines of symmetry'), cite evidence from their folding/mirror activity, and provide geometric reasoning. This strategy develops mathematical communication and connects the hands-on observation to formal geometric language, directly building toward the properties of reflection that will be formalised in Lesson 2.	Collect the 'I was right / I need to revise' annotation as an exit indicator. Specifically look for: (a) Whether learners now correctly identify 1 vertical line of symmetry for letter A (not 0, not 2). (b) Whether learners distinguish between a shape having a line of symmetry and 'looking nice' — evidence of conceptual rather than aesthetic understanding. (c) Whether at least 80% of learners can correctly state the number of lines for square, rectangle, equilateral triangle, and circle in the consensus table.
Driving Question Board (DQB)	Each learner receives two sticky notes (or tears a small piece of paper). On the first, they write one	1. Distribute sticky notes. Say: 'You have 2 minutes. Write your most burning question on the first	Driving Question Board with clustered question sorting. This metacognitive strategy makes	Review the 'sure about' notes: Are learners expressing genuine conceptual understanding ('A

Creation  VIDEO: Identifying Lines of Symmetry in Various Shapes Source: CK-12  Search ARES for similar videos  EXERCISE: Identify line symmetry Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	question the lesson has raised for them about mirrors, reflection, or symmetry. On the second, they write one thing they are now sure about. The teacher collects the 'question' notes and reads selected ones aloud before grouping them on the class Driving Question Board under three emerging clusters: 'About the IMAGE', 'About the DISTANCE', and 'About the FLIP/REVERSE'. Learners copy the three clusters and their own question into their exercise books as the start of their personal DQB tracker.	note — something about mirrors or reflection that you genuinely do not yet understand. On the second, write one thing you are now confident about after today's safari.' [WAIT 2 minutes — no talking.] 2. Collect question notes. Read 6–8 aloud without attribution. Say: 'Listen — do these questions sound familiar? Give a thumbs-up if you had a similar question.' 3. Model clustering on the DQB: 'Wanjiru asked: Why does the image appear the same distance behind the mirror as I am in front? That goes under DISTANCE. Otieno asked: Why is the letter A reversed left-right but not upside down? That goes under FLIP/REVERSE.' 4. Invite 2 learners to place remaining notes on the board. 5. Say: 'This board belongs to all of us. Every lesson in this unit, we will come back and answer some of these questions. By the end of the unit, every question on this board should have an answer.' 6. Point to the driving question: 'Why does a mirror show a perfect copy of the world — and where exactly does that copy live?' Write it at the top of the DQB. Say: 'That is the question we are all trying to answer together.'	learner thinking visible, creates shared intellectual ownership of the unit, and allows the teacher to diagnose which conceptual gaps are most prevalent. Clustering questions into 'IMAGE', 'DISTANCE', and 'FLIP' categories foreshadows the three key properties of reflection (congruence, equidistance, perpendicular bisector) that form the mathematical core of this sub-strand.	square has 4 lines of symmetry because it can be folded in 4 ways that match perfectly') or surface-level recall ('We used mirrors')? Review question notes: Are learners asking mathematical questions ('How do we calculate where the image is?') or only perceptual ones ('Why does my face look different in a mirror?')? The ratio of mathematical to perceptual questions indicates readiness to move into the properties of reflection in Lesson 2.
Model Building Phase  VIDEO: Determining Whether Various Shapes Have Lines of Symmetry Source: CK-12  Search ARES for similar videos	Learners draw their Initial Reflection Model in their exercise books. The model has three components: (A) A diagram — learners draw a simple plane figure (e.g., the letter 'A' or a Kanga-inspired shape) next to a vertical mirror line and sketch what they currently believe the image looks like, labelling 'object', 'mirror line', and 'image'. (B) A rule statement — learners complete the sentence: 'I think the mirror line	1. Say: 'Close your eyes for 10 seconds and picture Aisha standing in front of her mirror this morning. Picture the letter A next to the mirror at the front of this class. Now open your eyes and draw what you saw — but this time, add labels.' [WAIT 10 seconds with eyes closed.] 2. Give 4 minutes for individual model drawing. Circulate. Do not correct diagrams — only prompt: 'Have you labelled the mirror line?' /	Initial Model construction using the Predict-Observe-Explain-Model (POEM) cycle. Creating an explicit, labelled diagram forces learners to commit a mental model to paper, making misconceptions visible and revisable. The 'question flag' (starring the least confident part) builds scientific and mathematical humility and creates a personal anchor for revision in subsequent lessons. This model will be iteratively updated across the unit,	Scan models during circulation for three specific indicators: (a) Does the learner place the image behind the mirror (correct) or in front of it (misconception to address in Lesson 2)? (b) Does the learner draw the image the same size as the object, or larger/smaller (foreshadows understanding of congruence)? (c) Does the learner draw the image as laterally reversed (correct intuition from the letter A observation) or as an

 EXERCISE: Determine reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	works by ____.' (C) A question flag — learners star the one thing in their diagram they are least confident about. The teacher emphasises: 'This is your starting model. It does not have to be correct — it has to be honest. We will revise it every lesson until it is perfect.'	'Where do you think the image is — in front of the mirror or behind it?' [WAIT 10 seconds after each prompt.] 3. Cold-call 2 learners to hold up their models and describe them in one sentence each. Do NOT evaluate correctness publicly. Say: 'These are two honest starting models. We will revisit them next lesson.' 4. Say: 'On the back of your model page, write today's date and the words: INITIAL MODEL — Lesson 1. Leave space below — that is where you will write REVISED MODEL after Lesson 2.' 5. Close the lesson by returning to the phenomenon: 'Aisha is still puzzled. The driver on the matatu is still checking his mirrors. The letter A is still on that board. Next lesson, we will find the exact geometric rules that explain every single thing we saw today.'	providing a cumulative record of conceptual growth.	identical non-reversed copy? Document 3–4 models that show the most common misconception for targeted discussion in Lesson 2.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did the Symmetry Safari generate genuine curiosity, or did it feel procedural? If groups moved through the folding task without discussing, consider adding a 'justify your fold' requirement in future iterations. (2) How rich were the questions on the DQB? If most questions were perceptual ('Why does my face look different?') rather than mathematical ('How far behind the mirror is the image?'), consider seeding the DQB with one teacher-modelled mathematical question to raise the level of discourse. (3) Were learners able to distinguish between a square (4 lines) and a rectangle (2 lines)? This distinction is a known stumbling block — if more than 30% of groups recorded the wrong value for a rectangle, plan a brief 3-minute re-engagement at the start of Lesson 2 using a folded A4 sheet of paper (rectangular) to make the distinction concrete. (4) Did the Initial Model activity reveal the expected misconception that the image is in front of the mirror? If so, hold those 3–4 flagged models and use them (anonymously) as discussion anchors at the start of Lesson 2's Observe phase. (5) Was 40 minutes sufficient? The Observe Phase (Symmetry Safari) tends to run long if materials are not pre-cut. For future delivery, ensure all cut-outs are prepared before the lesson begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	A bold letter 'A' placed beside a plane mirror in the classroom produced a mirror image that appeared laterally reversed (flipped left-right). Leaves from the school compound, paper cut-outs of rectangles, squares, equilateral triangles, circles, and Kanga fabric swatches were folded and tested with a plane mirror. Different shapes yielded different numbers of lines of symmetry: circle (infinite), square (4), rectangle (2), equilateral triangle (3), scalene triangle (0).
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What did I learn?	A line of symmetry divides a plane figure into two congruent halves that are exact mirror images of each other. It is also called a mirror line. Shapes can have zero, one, more than one, or infinitely many lines of symmetry depending on their geometry. The fold test and the mirror test both confirm the same lines of symmetry, showing that folding and reflecting are geometrically equivalent operations.
How does this explain the phenomenon?	The mirror line acts as the line of symmetry — it is the invisible geometric rule that governs where the mirror image appears. When the letter 'A' is placed beside a mirror, the vertical line of symmetry of the letter aligns with the mirror line, producing a reversed image that is congruent to the original. The 'flip' observed in Aisha's bedroom mirror and in the matatu's rear-view mirror is a consequence of reflection across a mirror line, which reverses the figure laterally while preserving all distances and sizes. This is the foundational property that the rest of the unit will make precise and quantitative.

LESSON 2 (80 minutes (2 × 40-minute lessons)): Mirror, Mirror on the Wall — Properties of Reflection on a Plane Surface

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to discover and formalise the three geometric properties of reflection on a plane surface through hands-on measurement and peer discussion.
Knowledge	Learners will know that: (i) the image distance from the mirror line equals the object distance; (ii) the line joining an object to its image is perpendicular to the mirror line; (iii) the image is congruent to the object.
Skills	Learners will be able to: (a) determine the properties of reflection using objects and their images; (b) draw an image given an object and a mirror line on a plane surface; (c) measure and verify perpendicularity and equal distances using a ruler and set square.
Attitudes	Learners will appreciate the role of reflection in real-life situations and take responsibility for instruments used during construction activities.
Key Inquiry Question	How is reflection important in day-to-day activities?
Purpose in Storyline	In Lesson 1 learners identified lines of symmetry and posed questions about the mirror behaviour they observe daily — in the bedroom mirror, the matatu side mirrors, and the letter 'A' demonstration. Lesson 2 moves from noticing to measuring: learners stand at known distances from a plane mirror to feel the phenomenon, then use ruler, set square, and plain paper to locate image points precisely and extract the three governing properties. This evidence becomes the geometric foundation the class will rely on for all subsequent Cartesian-plane and real-life reflection tasks.
Safety Notes	Handle plane mirrors carefully — edges may be sharp. Place mirrors flat on desks when not in use. Learners with photosensitive conditions should be seated away from direct reflected sunlight.

B. LESSON OVERVIEW

Learners begin by predicting what will happen to their image distance when they step closer to or further from a plane mirror. They then gather evidence in two phases: (1) a physical mirror activity where they stand at measured distances and record observations, and (2) a paper-based construction activity using a ruler and set square to locate image points for objects placed at various distances from a drawn mirror line. From their measurements learners identify and formalise the three properties of reflection. They connect every finding back to the anchoring phenomenon — the bedroom mirror, matatu rear-view mirrors, and the letter 'A' — and update the Driving Question Board with new evidence. The lesson closes with a cumulative model sketch that annotates the three properties geometrically.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Geometric constructions: perpendicular	Learners receive a half-page prediction slip showing a top-view diagram of a girl (labelled 'Aisha') standing 30 cm in front of a plane mirror. Three questions are	Distribute prediction slips face-down. Say: 'Before we touch any equipment today, I want to know what your brain already thinks. Turn over your slip — you have 4	Predict-Then-Investigate (Elicit Prior Knowledge): surfacing predictions before evidence gathering activates schema, creates cognitive dissonance when	Teacher scans slips as learners write and notes: (a) what fraction predict 'same distance' (correct); (b) whether learners draw the correct hand being raised

bisector Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12 Search ARES for similar readings	printed: (i) If Aisha moves to 60 cm from the mirror, where will her image appear — closer, the same distance behind, or further behind the mirror? Mark your prediction with an X. (ii) If she raises her RIGHT hand, which hand does the image raise? Draw an arrow. (iii) Write one sentence: what invisible rule do you think determines exactly where the image lands? Learners work individually and silently for 4 minutes, then share predictions with a shoulder partner for 2 minutes. The teacher takes a quick show-of-hands poll for each option on question (i) and records the tally visibly on the board under the heading 'Our Predictions — Lesson 2'.	minutes, working alone and in silence.' Set a visible timer. After 4 minutes: 'Turn to your shoulder partner. Compare your predictions. Do you agree? You have 2 minutes.' After partner talk, conduct the poll: 'Hands up if you predicted the image moves closer.' Tally. 'Hands up — same distance.' Tally. 'Further away.' Tally. Cold-call 2 learners from the largest group and 1 from a minority group: 'Baraka, tell us the reasoning behind your prediction.' WAIT 12 seconds. Record their reasoning verbatim in a second column beside the tally labelled 'Our Reasoning'. Say: 'We are going to put every one of these predictions to the test right now. Keep your slip — you will revisit it at the end of class.'	results differ, and gives learners a personal stake in the upcoming measurements. Recording reasoning publicly makes thinking visible and gives the teacher diagnostic information about misconceptions (e.g. 'the image gets smaller when you move away').	(common error: same hand). Misconceptions about image size changing with distance flag learners who may need targeted questioning during the observation phase. Tally on board becomes a running record the class will verify.
Observe Phase  VIDEO: Geometric constructions: perpendicular bisector Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12 Search ARES for similar readings	PART A — Physical Mirror Activity (10 min): Groups of four receive one plane mirror (≥ 20 cm tall), a 1-metre tape measure, and a recording table with columns: Object Distance from Mirror (cm) Estimated Image Distance from Mirror (cm) Same / Closer / Further. One learner stands at 20 cm, 40 cm, 60 cm, and 80 cm from the mirror (marked with masking-tape strips on the floor). A second learner peers into the mirror from the side at eye level and calls out where the image appears to be. The group records. Roles rotate so each learner stands in front of the mirror at least once. Learners note in the final column whether the image appears to be the same size each time. PART B — Paper Construction Activity (15 min): Each learner receives a sheet of plain A4 paper, a ruler, and a set square. They draw a vertical mirror	PART A — Before groups begin, hold up the mirror: 'This is the same type of mirror Aisha uses in her bedroom and the same principle behind the matatu driver's rear-view mirror. We are going to measure what it actually does.' Assign roles aloud. Circulate; after 3 minutes prompt: 'Has anyone noticed a pattern yet? Don't announce it — write it in your table.' Cold-call one group at the 8-minute mark: 'Zawadi, what does your table show so far?' WAIT 10 seconds. Affirm data, not conclusions yet: 'Interesting data — hold your conclusion until Part B confirms it.' PART B — Model the first construction step on the board with an oversized compass and ruler: 'I draw a perpendicular from P_1 to M — that means a 90-degree angle, confirmed with the set square like this.' Demonstrate set-square placement. Then step	Data-Collection-and-Pattern-Detection (Structured Observation): the two-part sequence — physical mirror first, then precise paper construction — moves learners from qualitative noticing ('the image seems the same distance away') to quantitative confirmation (measured cm values). This mirrors the NGSS practice of Planning and Carrying Out Investigations. The structured recording table prevents premature closure and ensures every learner has personal data to reason from in the Explain phase.	Observable indicators: (1) Set-square held correctly against M — teacher physically checks 100% of pairs during Part B. (2) Object distance and image distance columns show equal values (± 1 mm tolerance acceptable) — if a learner's table shows discrepancies > 2 mm, teacher asks 'Did you extend the perpendicular exactly the same length on both sides? Show me how you measured.' (3) Angle column shows 90° — any value far from 90° indicates the perpendicular was not constructed correctly. (4) Triangle sides comparison shows equality — confirms congruence property.

	line M down the centre. They place an object point P_1 at 3 cm to the left of M, P_2 at 5 cm to the left, P_3 at 7 cm to the left, and P_4 at 4 cm to the left but 3 cm higher. For each point they: (i) draw a perpendicular from the point to M; (ii) extend it an equal distance on the other side; (iii) mark the image point P_1', P_2', P_3', P_4'. They then connect $P_1P_2P_3$ to form a triangle and connect the corresponding image points to form the image triangle. Learners measure: object distance vs. image distance for each point; the angle between PP' line and M; and the sides of the object triangle vs. image triangle. They record all measurements in a table with columns: Point Object Distance (cm) Image Distance (cm) Angle PP' makes with M ($^\circ$) Are they equal?	back: 'Now you complete P_1 through P_4. I will circulate and ask you to show me your set square is flush against M.' Circulate and check every pair. At the 12-minute mark ask the class to freeze: 'Everyone measure the angle that the line P_1P_1' makes with M and write it. I am going to cold-call three people.' WAIT 12 seconds. Cold-call 3 learners across the room. Record responses on the board. Ask: 'What do all three answers have in common?' WAIT 10 seconds. Then: 'Now measure object distance vs. image distance for each point and compare the sides of your two triangles.'		
Explain Phase  VIDEO: Reflecting shapes: diagonal line of reflection <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar	Learners examine their completed tables from Part B and, working in pairs, draft three 'property statements' in their own words. After 5 minutes, the class participates in a Numbered-Heads-Together activity: each group of four numbers off 1–4. The teacher calls a number; that learner shares one property. The class evaluates each statement against their own data and refines the wording collaboratively until consensus is reached. The teacher scribes the agreed formal statements on the board: PROPERTY 1 — The image distance from the mirror line equals the object distance from the mirror line (object distance = image distance). PROPERTY 2 — The line joining any object point to its corresponding image point is perpendicular to the mirror line. PROPERTY 3 — The image is	After pairs draft their property statements: 'I want each pair to have three statements written. You have 5 minutes. Go.' WAIT — circulate silently reading drafts. After 5 minutes: 'Numbers-Heads-Together — number off in your group now: 1, 2, 3, 4.' Pause. 'All number-2s — stand up. Number-2 from Group A — Fatuma — read Property 1 as your group wrote it.' WAIT 12 seconds. Write Fatuma's words verbatim on the board. 'Class, look at your own data. Does Fatuma's statement match what you measured? Thumbs up, thumbs sideways, or thumbs down.' Address thumbs-sideways: 'Hamisi, what part doesn't match your data?' WAIT 10 seconds. Refine the statement live. Repeat for Properties 2 and 3 with number-3 and number-4 learners. Once all three are formalised,	Numbered-Heads-Together with Collaborative Refinement (NGSS Constructing Explanations): by requiring learners to draft statements before hearing correct wording, this strategy ensures the formal properties emerge from their own data rather than from teacher transmission. The public refinement process models mathematical argumentation — a core mathematical process skill in the KICD curriculum — and gives every learner ownership of the final statements.	During the Numbered-Heads-Together round, teacher listens for: (1) Use of the word 'perpendicular' — absence suggests the concept needs re-teaching; (2) Conflation of 'congruent' with 'identical' without noting lateral reversal — probe with 'Is the image exactly the same, or is anything different?'; (3) Correct calculation of 2.4 m total distance in the Otieno exercise — errors here often indicate learners added image distance to mirror thickness rather than to object distance. Quick written check: teacher asks every learner to write one sentence answering 'Which property explains why the letter A looks flipped in a mirror?' — collected as an exit ticket for this phase.

readings	congruent to the object (same size and shape, laterally reversed). Learners copy the three formalised properties into their notebooks under the heading '2.2 Reflection — Properties'. They then apply the properties immediately to two quick exercises: (a) Given a mirror line and a point 4.5 cm from it, locate and label the image point; (b) Otieno stands 1.2 m from a plane mirror at Nairobi's Railways Museum hall of mirrors — how far is his image from the mirror, and what is the total distance between Otieno and his image?	cold-call for the quick exercises: 'Exercise (b) — Otieno and the Railways Museum mirror. Who can set up the calculation? I need someone who hasn't spoken yet.' WAIT 12 seconds. 'Chebet — go ahead.' Confirm: image is 1.2 m behind mirror; total distance = $1.2 + 1.2 = 2.4$ m. Explicitly connect: 'Now think back to Aisha this morning. Her image in her bedroom mirror is the same distance behind the glass as she is in front. Which property explains that?' WAIT 10 seconds. Cold-call 1 learner.		
Driving Question Board (DQB) Creation  VIDEO: Reflecting points in the coordinate plane <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Justify constructions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives two sticky notes — one green, one orange. On the GREEN note they write one thing they can now answer about the driving question ('Why does a mirror show a perfect copy of the world — and where exactly does that copy live?'). On the ORANGE note they write one new question that today's evidence raised but has not yet answered. Learners post their notes on the class DQB, which is divided into three columns: 'What We Predicted', 'What Evidence Showed', 'New Questions'. The teacher reads three orange notes aloud and the class votes (hands) on which question is most important to investigate next. The winning question is circled and dated as 'Most Urgent — Lesson 3 target'.	Distribute sticky notes. 'You have 3 minutes — GREEN: one thing our measurements today let you answer about the driving question. ORANGE: one new question our measurements raised. Write in full sentences.' After 3 minutes: 'Post green notes in the 'What Evidence Showed' column and orange notes in 'New Questions'.' Circulate to read notes as learners post. Select 3 orange notes that represent different kinds of questions (conceptual, procedural, application). Read each aloud: 'Kamau asks: Does this work the same way if the mirror line is slanted — like a tilted mirror in a barbershop?' Read a second. Read a third. 'Hands up for the question you most want answered in our next lesson.' Tally. Circle the winner. 'We will track this question all the way to Lesson 8. Write it in your notebook under our Driving Question.' Explicitly retrieve predictions from the slip: 'Look at your prediction slip. Put a tick next to each prediction today's evidence confirmed, and a cross next to each it disproved. You do	Driving Question Board — Evidence Mapping (NGSS Engaging in Argument from Evidence): the DQB makes the cumulative inquiry arc visible and habituates learners to distinguishing 'what we know from evidence' from 'what we still wonder'. The prediction-slip revisit creates a metacognitive moment — learners experience the productive discomfort of having predictions corrected by data, which deepens understanding more than being told the correct answer at the outset.	Teacher reads all green sticky notes after class to assess whether learners can articulate the three properties in their own words and connect them to the driving question. A green note that says only 'the image is behind the mirror' without reference to distance or perpendicularity signals incomplete understanding. Orange notes are analysed for scope: questions about curved mirrors, Cartesian-plane mirror lines, or lateral reversal indicate readiness for extension; questions about basic measurement procedure indicate learners who need consolidation before Lesson 3.

		not need to hand it in — it is your personal evidence record.'		
Model Building Phase  VIDEO: Determining reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflection Symmetry (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners open to their cumulative model page (started in Lesson 1 with a sketch of Aisha and her bedroom mirror). They add a top-view diagram of the bedroom scene labelled with: (i) a double-headed arrow between Aisha and the mirror line annotated 'd_object'; (ii) a matching double-headed arrow on the image side annotated 'd_image = d_object'; (iii) a dashed line joining Aisha to her image labelled '90°' where it crosses the mirror line; (iv) a small congruence symbol ($\cong$) between Aisha's figure and the image figure. In a text box beside the diagram they write: 'My model now explains: the image appears as far behind the mirror as I stand in front, because Property 1 says object distance = image distance. It is the same size because the image is congruent (Property 3). The image appears directly across the mirror because the joining line is perpendicular (Property 2).' Below this they write: 'My model cannot yet explain: [student fills in — e.g., what happens when the mirror line is tilted, or how to find the image on a coordinate grid].' The final 3 minutes are used for a partner gallery walk: partners swap model pages, read each other's annotations, and write one specific 'I notice...' comment on a sticky note attached to the page.	Project the Lesson 1 model sketch on the board (photograph taken at the end of Lesson 1). 'Here is where our model was at the end of last lesson — we knew mirrors existed and we had questions. Today we have evidence. Open your model page.' Walk through the labelling step by step: 'First, draw the double-headed arrow for object distance and label it d_object. I'll wait while you do that.' WAIT 20 seconds. 'Now mirror it on the image side, d_image, and write the equation. Set squares away — just annotate.' WAIT 20 seconds. 'Add the 90-degree label at the crossing point.' WAIT 15 seconds. 'Congruence symbol between Aisha and her image.' WAIT 15 seconds. 'Now the text box — 3 sentences: what my model explains, which property it uses, what it still can't explain.' WAIT 3 minutes. Circulate and read text boxes; stamp or initial completed models. For the gallery walk: 'Swap pages with your partner. Read carefully. Write one 'I notice...' on their page — something specific about their diagram or explanation.' WAIT 2 minutes. 'Return pages.' Close: 'By Lesson 8 this model will be able to explain everything we see in Aisha's mirror, in the matatu's mirrors, and in the letter-A demonstration. Today you added the three pillars. Hold on to this model — it grows with us.'	Cumulative Model Revision (NGSS Developing and Using Models): requiring learners to annotate the same diagram across multiple lessons externalises the growth of their understanding and prevents fragmented knowledge. The 'what my model cannot yet explain' prompt is deliberate — it sustains productive epistemic humility and keeps the inquiry arc open, motivating subsequent lessons. The partner gallery walk introduces peer critique as a mathematical practice.	Teacher checks model pages for three specific indicators: (1) Both distance arrows labelled and equation $d_{\text{object}} = d_{\text{image}}$ written — if missing, learner did not internalise Property 1; (2) '90°' correctly placed at the intersection, not elsewhere on the diagram — misplacement indicates confusion between the perpendicular property and the mirror line itself; (3) Congruence symbol present and the text box explicitly names all three properties — absence of any property in the text box flags a gap. Models are photographed for the teacher's record and returned with a written question prompt tailored to each learner's identified gap, ready for the start of Lesson 3.

D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) During the Predict Phase, what was the most common misconception on prediction slips — did learners think the image size changes with distance? How will you address this in Lesson 3? (2) In Part B of the Observe Phase, were learners able to use the set square correctly to construct perpendiculars, or was the technique a barrier that slowed the investigation? If so, consider a 3-minute set-square drill at the start of Lesson 3. (3) Did the Numbers-Heads-Together produce all three properties, or did Property 2 (perpendicularity) consistently fail to emerge? If learners kept conflating 'opposite side' with 'perpendicular', plan a targeted re-teach using the analogy of the road crossing the river at right angles — a context familiar from Kenyan geography lessons. (4) Review all green sticky notes: what percentage articulated the connection between the measured data and the anchoring phenomenon (Aisha's mirror, matatu mirrors, letter A)? If fewer than 70% made the connection explicitly, open Lesson 3 by cold-calling learners to make that connection before moving to Cartesian-plane reflection. (5) Which orange DQB question won the vote? Ensure your Lesson 3 plan directly addresses it in the opening Predict phase.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners stood at measured distances (20 cm, 40 cm, 60 cm, 80 cm) from a plane mirror and recorded the apparent position of their image. They then constructed image points for four object points on plain paper using a ruler and set square, measuring object distance, image distance, and the angle the joining line made with the mirror line for each point. They also compared the sides of an object triangle with those of its image triangle.
What did I learn?	From their measurements learners discovered and formalised three properties of reflection on a plane surface: (1) object distance from the mirror line equals image distance from the mirror line; (2) the line joining an object point to its image is perpendicular to the mirror line; (3) the image is congruent to the object. They applied these to calculate Otieno's total distance from his image (2.4 m) in a Kenyan context and annotated Aisha's bedroom mirror scenario with all three properties.
How does this explain the phenomenon?	The three formalised properties explain why Aisha's bedroom mirror shows her image the same distance behind the glass as she stands in front (Property 1), why the image appears directly 'across' the mirror rather than at an angle (Property 2), and why every detail — the Kanga pattern, the button count, the letter A — is preserved in size and shape even though it is laterally reversed (Property 3). The matatu driver's rear-view mirror follows the same three rules, so vehicles behind appear at a true proportional distance.

LESSON 3 (40 minutes): Drawing Images on a Plane Surface — Construction Practice

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To develop learners' ability to accurately construct reflected images of plane figures across horizontal, vertical, and oblique mirror lines using a geometrical set, and to annotate constructions with object distance, image distance, and perpendicular bisectors.
Knowledge	Learners will know that: (i) the image of a point under reflection lies on the perpendicular from the point to the mirror line, at an equal distance on the opposite side; (ii) this rule holds for any orientation of the mirror line — horizontal, vertical, or oblique; (iii) the image of a plane figure is congruent to the object and laterally reversed.
Skills	Learners will be able to: (i) use a ruler and set square to drop a perpendicular from each vertex of a plane figure to the mirror line; (ii) measure and mark the image vertex at the same distance on the opposite side; (iii) join image vertices to construct the complete reflected figure; (iv) label object distance (d_o), image distance (d_i), and the perpendicular bisector for each corresponding pair of points.
Attitudes	Learners will appreciate the precision and care required in geometric construction, recognising that the neat, rule-governed behaviour of mirror images — as seen in the morning bedroom mirror, the matatu side mirrors, and the teacher's letter 'A' demonstration — is a direct consequence of the exact geometric properties they are constructing by hand.
Key Inquiry Question	How is reflection important in day-to-day activities?
Purpose in Storyline	In Lessons 1 and 2 learners identified lines of symmetry and discovered the properties of reflection (equal distances, perpendicularity, congruence, lateral reversal) through physical objects and plane mirrors. Lesson 3 now demands that learners operationalise those properties: using only a geometrical set, they must construct — not just observe — the precise location of a reflected image. This bridges observation and formal geometric reasoning, and directly answers a growing Driving Question Board question: 'What invisible rule governs exactly where the image lands?'
Safety Notes	Compasses have sharp points — learners must keep compass points directed downward onto the paper and must not wave instruments. Rulers and set squares should be placed flat on desks when not in use. Learners working in pairs should agree on who holds the ruler before the other marks, to avoid slipping instruments.

B. LESSON OVERVIEW

Learners use rulers, set squares, and compasses from their geometrical sets to construct reflected images of triangles, quadrilaterals, and an irregular polygon across three mirror-line orientations (horizontal, vertical, oblique). The teacher models each step with a think-aloud on the board, narrating every geometric decision. Learners then practise in pairs on plain paper, annotating each construction with object distance (d_o), image distance (d_i), and the perpendicular bisector of each corresponding pair of vertices. The lesson is anchored throughout in the phenomenon: each construction is framed as "building the invisible geometric rule that explains why Aisha's reflection in her bedroom mirror appears exactly as far behind the glass as she stands in front of it, and why the letter 'A' the teacher held up was the same size but perfectly flipped."

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Reflecting shapes: diagonal line of reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners receive a plain-paper worksheet showing a scalene triangle ABC and a vertical mirror line m drawn 4 cm to its right. Without any instruments, they sketch — freehand — where they predict the reflected image A'B'C' will appear, and jot a one-sentence rule they are using to guide their sketch. They share predictions with a neighbour for 90 seconds.	Distribute worksheets face-down. Say: 'Before we touch our geometrical sets today, I want to see what your brain already knows. Turn your sheet over — you have 90 seconds to sketch, freehand, where you think the mirror image of triangle ABC will appear across line m. Write one sentence explaining your rule. Go.' Set a visible timer for 90 seconds. Circulate silently — do NOT correct. After time, say: 'Turn to your neighbour. In 30 seconds, tell them one thing your sketches agree on and one thing that is different.' Cold-call 3 pairs: 'Kamau and Njeri — what did you agree on?' / 'Akinyi and Omondi — where did your images differ?' / 'Wanjiku and Mwangi — what rule did you both write down?' Allow 10 seconds of wait time after each question before accepting responses. Record 2–3 student-stated rules on the board under the heading 'Our Predictions'. Do not confirm or deny — say: 'Let's find out which of these rules survives today's constructions.'	Predict-then-Compare: by committing a freehand prediction to paper before instruction, learners activate and expose their current geometric mental models. The public posting of student-stated rules creates cognitive stakes — learners are motivated to test their own claim during construction.	Teacher scans sketches during circulation for two observable indicators: (1) Has the learner placed the image on the correct side of the mirror line (opposite to the object)? (2) Does the image appear roughly the same size as the object? Learners who place the image on the same side, or dramatically resize it, are noted for targeted support during the Observe phase. Teacher does not intervene — data is for grouping decisions only.
Observe Phase  VIDEO: Geometric constructions: perpendicular bisector Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections Source: Khan Academy (English)	The teacher performs a slow, narrated step-by-step construction of the reflection of triangle ABC across the vertical mirror line on the board (large format, using a metre ruler and chalk set square). Learners watch and annotate a printed 'construction guide' card that lists each step as a blank to be completed. After the modelled triangle, the teacher constructs a second example — a rectangle across a horizontal mirror line — at increased pace, narrating less. Learners then open their geometrical sets and replicate both constructions on plain paper. They	Stand at the board with a large-format version of the worksheet. Say: 'Watch every move I make — I am going to think out loud so you hear the geometry, not just see the pencil move.' Step 1 — say: 'I choose vertex A. I need a line from A that hits mirror line m at exactly 90 degrees. I place my set square with the right-angle corner on m and slide it until the vertical edge passes through A.' Draw the perpendicular. Step 2 — say: 'I measure the distance from A to m along this perpendicular — I get 3 cm. I mark a point A' exactly 3 cm on the OTHER side of m, along	Modelled Think-Aloud followed by Immediate Replication: the teacher narrates the geometric reasoning at each construction step (not just the procedure), so learners hear the 'why' (equal distances, perpendicularity) before they practise the 'how'. Immediate replication on plain paper consolidates procedural fluency while the reasoning is still audible in working memory. The deliberate progression — vertical, then horizontal, then oblique — builds generalisation: learners recognise that the same perpendicular-bisector rule governs all mirror-line	Teacher uses a 3-point circulating checklist per pair: (1) Perpendicular dropped correctly from each vertex to the mirror line — yes/no; (2) $d_o = d_i$ measured and confirmed — yes/no; (3) Image vertices joined to form a closed, congruent figure — yes/no. Any pair scoring 0–1 receives immediate side-by-side reteaching of Step 1. Any pair scoring 2 is prompted: 'Measure d_o and d_i with your ruler — read the numbers aloud to each other.' Pairs scoring 3 proceed to the oblique challenge task.

- US curriculum) Search ARES for similar readings	annotate: label each vertex pair (e.g. $A \rightarrow A'$), draw and label the perpendicular from A to m, mark d_o and d_i with a double-headed arrow, and highlight the perpendicular bisector. Finally, each learner attempts an oblique (45°) mirror line with an irregular quadrilateral — a shape chosen because it cannot be 'guessed' by symmetry intuition alone.	the same perpendicular.' Mark A'. Step 3 — repeat for B and C. Step 4 — say: 'I join $A'B'C'$. I now label d_o and d_i with arrows, and I mark the perpendicular bisector.' After completing triangle: pause and ask — 'Why must d_o equal d_i? Think for 10 seconds.' [Wait 10 seconds.] Cold-call 2 learners. Then say: 'Recall Aisha standing in front of her bedroom mirror this morning — she noticed her image appeared the same distance behind the glass as she stood in front. She just observed what you are now constructing.' Proceed to model the rectangle across the horizontal line at faster pace. Release to pair practice — circulate and ask targeted questions: 'Show me the perpendicular you dropped from this vertex.' / 'Is d_o equal to d_i — measure and confirm.' / 'Your image vertex is 1 mm off — which step introduced the error?' For the oblique line task, say: 'The mirror line is now at 45° — your set square has a 45° angle. Which edge will you align to m to drop your perpendicular?' Wait 10 seconds before cold-calling.	orientations, just as the same geometry governs both Aisha's bedroom mirror and the matatu's angled side mirror.	
Explain Phase  VIDEO: Reflecting shapes: diagonal line of reflection Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Determine reflections (advanced) <i>Related topic: not an exact match</i>	Each pair receives a 'What went wrong?' card showing three common construction errors (image placed on same side as object; image vertex not at equal distance; lines joining image vertices not straight). Pairs identify which error matches a deliberately flawed construction printed on the card, write a one-sentence correction using geometric language (perpendicular, equidistant, bisector), and then revisit their own completed constructions to check for the same errors. The class	Distribute 'What went wrong?' cards. Say: 'You have 3 minutes — identify the error on the card, write your correction sentence using at least two of these words: perpendicular, equidistant, bisector. Go.' Circulate. After 3 minutes, cold-call 3 pairs to read their correction sentence. After each, ask the class: 'Agree or refine? Thumbs up if you agree.' If thumbs are mixed, say: 'Omond, what would you change in that sentence?' [Wait 10 seconds.] Then return to the board predictions posted in Phase 1.	Error Analysis (Worked Example Critique): presenting pre-made errors externalises common misconceptions without targeting any individual learner. Correcting a flawed example requires learners to articulate the correct property, which deepens understanding beyond procedural recall. Connecting the corrected rule back to the phenomenon ties formal geometric language to the original observable puzzle, advancing the driving question.	Collect correction sentences from three cold-called pairs. Observable indicator of understanding: the sentence must contain (i) a reference to perpendicularity, (ii) a statement of equal distances, and (iii) correct identification of which side the image appears on. Sentences that omit any of the three are flagged — the teacher notes those learners for re-engagement in Lesson 4's Cartesian-plane work.

for this lesson. Source: Khan Academy (English - US curriculum) Search ARES for similar readings	reassembles for a whole-group discussion linking construction steps to the reflection properties stated in Lesson 2. Learners add a 'Construction Rule' box to their notes: 'For any point P at distance d from mirror line m, image P' lies on the perpendicular from P to m, at distance d on the opposite side — so PP' is bisected perpendicularly by m.'	Say: 'Look at the rules we wrote at the start. Kwame wrote: "the image goes on the other side and is the same size." Is that now complete enough, or can we make it more precise?' Invite a volunteer to rewrite the rule using today's construction language. Anchor explicitly to phenomenon: 'When the maths teacher held up the letter A next to the plane mirror, the image A appeared perfectly preserved — same size, same shape. You have now constructed the geometric reason WHY. The perpendicular bisector property is the invisible rule Aisha and the class were puzzled by.' Dictate the Construction Rule box slowly for learners to copy.		
Driving Question Board (DQB) Creation  VIDEO: Geometric constructions: perpendicular bisector Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Each learner writes on a sticky note (or a paper strip if sticky notes are unavailable): one thing today's construction confirmed about the mirror phenomenon, and one new question it raised. They post contributions on the class Driving Question Board under two columns: 'We can now explain...' and 'We still wonder...'. The class does a 2-minute silent gallery walk of new postings, then the teacher reads two 'still wonder' questions aloud for the class to decide: 'Is this answered by what we already know, or does it need future investigation?'	Say: 'Take a sticky note. On one side write: one thing today's construction proves about why Aisha's image is exactly as far behind her mirror as she stands in front. On the other side write: one question today raised that we haven't answered yet. You have 90 seconds.' Circulate — prompt stuck learners: 'Think about the oblique mirror line task — did anything surprise you?' After posting, read two 'still wonder' notes aloud — e.g. 'Does the rule still work if the mirror line is curved?' / 'What happens when two mirror lines cross?' Say to class: 'Hands up — answered already, or needs future lessons?' Tally publicly. Say: 'The question about what happens when the mirror line is on a coordinate grid — that is exactly where we go next.' Leave both columns visible for the remainder of the unit.	Driving Question Board — Progressive Disclosure: the DQB functions as a public, cumulative knowledge-tracking tool. Separating 'we can explain' from 'we still wonder' makes the boundary of current understanding visible to both learners and teacher. Learner-generated questions in the 'still wonder' column become the motivational bridge to Lesson 4 (reflection on a Cartesian plane), preserving the narrative of the phenomenon rather than treating lessons as isolated topics.	Teacher reads all posted 'we can now explain' notes before the next lesson. Observable success indicator: the note references at least one of — perpendicular bisector, equal distances, or congruent image — in connection to the phenomenon (mirror / Aisha / letter A). Notes that are purely procedural ('we draw a perpendicular') without linking to the phenomenon indicate surface-level engagement and flag those learners for a targeted question at the start of Lesson 4.
Model	Learners return to their cumulative	Say: 'Open your Reflection Model	Cumulative Model Revision: the	Teacher collects models at the end

Building Phase  VIDEO: Determining reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Justify constructions <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	'Reflection Model' — a large annotated diagram they began in Lesson 1 and updated in Lesson 2. Today they add a new panel labelled 'Lesson 3 — Construction Evidence'. In this panel they paste or redraw their best construction from today (the oblique mirror line with the irregular quadrilateral), annotate it fully (d_o, d_i, perpendicular bisector, vertex labels), and write two sentences: (1) what this construction proves about the location of the image, and (2) how this explains one specific detail from the morning phenomenon (Aisha's mirror, the matatu side mirror, or the letter A). Pairs share their two sentences with another pair and suggest one improvement to each other's wording.	from Lessons 1 and 2. You are adding today's construction evidence. Choose your clearest construction — the oblique mirror line — and paste or redraw it carefully in the new panel. Annotate completely: d_o, d_i, perpendicular bisector, all vertices. Then write your two sentences. You have 5 minutes.' Circulate and read sentences as they are written. Prompt with: 'Your first sentence says the image is the same distance away — can you add the word perpendicular to make that more precise?' / 'Your second sentence mentions Aisha — can you say exactly which property of your construction explains what she saw?' After 5 minutes: 'Share your two sentences with the pair next to you. Give them one improvement suggestion — something they could add or make more precise. 2 minutes.' Cold-call one pair to read their final second sentence to the class. Close with: 'Your model is growing. In Lesson 4 we move the mirror line onto a coordinate grid — and your model will need a new panel to show what changes and what stays exactly the same.'	multi-lesson annotated diagram externalises learners' growing conceptual model of reflection, making prior knowledge visible and revisable. Requiring learners to connect the new construction panel explicitly to the phenomenon prevents the model from becoming a mere collection of diagrams — each addition must advance the explanation of the driving question. Peer sentence-sharing introduces low-stakes critique that sharpens geometric language use.	of the lesson and checks two indicators before Lesson 4: (1) Is the oblique-mirror construction correctly annotated with $d_o = d_i$ confirmed and perpendicular bisectors drawn? (2) Do the two written sentences correctly link a geometric property (perpendicularity / equal distance / congruence) to a specific observable detail from the phenomenon? Models meeting both indicators are ready for the Cartesian-plane extension. Models meeting only one indicator receive a written prompt: 'Add the missing property — which measurement or line is not yet labelled?' Models meeting neither indicator trigger a brief one-to-one review at the start of Lesson 4.
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D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Construction accuracy — what proportion of pairs correctly achieved $d_o = d_i$ and a properly dropped perpendicular on the oblique mirror line? If fewer than two-thirds succeeded, Lesson 4 should open with a 5-minute re-modelling of the oblique case before introducing the Cartesian grid. (2) Language precision — did learners use 'perpendicular bisector', 'equidistant', and 'congruent' spontaneously in the Explain phase, or only when prompted? If prompted, build a vocabulary warm-up into Lesson 4. (3) DQB quality — do the 'still wonder' questions show genuine curiosity about the Cartesian plane, or are they still at the level of 'why does the image flip?' (a Lesson 2 question)? If the latter, the phenomenon connection may need strengthening in the Lesson 4 hook. (4) Equity check — were the same 3–4 learners cold-called throughout? Review cold-call notes and ensure different learners are targeted in Lesson 4. (5) Time management — was there sufficient time for the full oblique-line construction, or was it rushed? If rushed, consider whether the 'What went wrong?' card in Phase 3 can be set as a homework task to recover time, given that plain-paper plain-language construction is the core Grade 10 KICD skill for this sub-strand.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners watched the teacher construct the reflected image of a triangle and a rectangle across vertical and horizontal mirror lines, narrating each geometric step (perpendicular from vertex to mirror line; measuring and marking equal distance on the opposite side; joining image vertices). They then observed their own constructions take shape on plain paper using rulers and set squares.
What did I learn?	The image of any point P under reflection lies on the perpendicular from P to the mirror line m, at a distance equal to the distance from P to m — on the opposite side. This means m is the perpendicular bisector of the segment joining any object point to its image. This rule holds regardless of whether m is horizontal, vertical, or oblique, and explains precisely why Aisha's image appears exactly as far behind her bedroom mirror as she stands in front of it.
How does this explain the phenomenon?	Using the perpendicular-bisector construction rule, learners can now explain: (1) why the image is the same size as the object (congruence is preserved because every vertex is mapped by the same equal-distance rule); (2) why the image appears 'behind' the mirror at the same distance as the object is 'in front' ($d_o = d_i$ by construction); and (3) why the image is laterally reversed (the perpendicular mapping flips the left–right orientation). These three facts together answer the core puzzle of the phenomenon — the driving question 'where exactly does the copy live?' now has a geometric construction answer.

LESSON 4 (40 minutes): Reflection on the Cartesian Plane I — Mirror Lines Along the Axes ($x = 0$ and $y = 0$)

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners move the mirror line from a physical surface onto a Cartesian grid and discover the exact coordinate transformation rules for reflection in the y-axis ($x = 0$) and the x-axis ($y = 0$), directly explaining why Aisha's image appears laterally reversed yet the same size and distance behind her mirror.
Knowledge	– State the coordinate rule for reflection in the y-axis: $(x, y) \rightarrow (-x, y)$ – State the coordinate rule for reflection in the x-axis: $(x, y) \rightarrow (x, -y)$ – Explain that the mirror line ($x = 0$ or $y = 0$) is the perpendicular bisector of the segment joining each object point to its image point
Skills	– Plot object vertices and their reflected images accurately on a Cartesian plane given the coordinate rules for $x = 0$ and $y = 0$ – Verify reflections by checking that the mirror line is equidistant from and perpendicular to each object–image pair – Derive the coordinate rule by comparing object and image coordinates in a table
Attitudes	– Demonstrate precision and care when plotting points and labelling coordinates on a Cartesian plane – Appreciate that abstract coordinate rules are rooted in the same physical mirror properties encountered in everyday Kenyan life
Key Inquiry Question	How is reflection important in day-to-day activities?
Purpose in Storyline	In Lessons 1–3 learners established the three physical properties of reflection and practised constructing images on a plane surface. Lesson 4 relocates the mirror line onto the Cartesian plane: the y-axis becomes Aisha's bedroom wall mirror and the x-axis becomes the floor mirror in the matatu, letting learners derive precise algebraic rules that will power all subsequent coordinate-reflection work and ultimately answer why the image appears exactly as far behind the glass as the object is in front.
Safety Notes	No specific hazards. Ensure sharp pencils are handled carefully during construction work. Rulers should be passed — not thrown — between partners.

B. LESSON OVERVIEW

This lesson bridges the physical mirror work of Lessons 1–3 with the algebraic precision of coordinate geometry. Learners begin by predicting what happens to the coordinates of a point when it is reflected in a vertical or horizontal mirror, drawing on what they already know: the image is the same distance behind the mirror as the object is in front. They then observe the pattern by plotting triangles and quadrilaterals that represent landmarks on a simplified grid map of a school compound in Nairobi, reflecting them in both axes, and recording a systematic table of object and image coordinates. The pattern in the table leads them to state the two coordinate rules explicitly.

In the Explain phase learners apply the rules to new shapes — including a quadrilateral whose vertices are labelled with distances matching the matatu rear-view mirror scenario from the anchoring phenomenon — and verify each image by the three properties established in Lesson 2 (equal distance, perpendicularity, congruence). The teacher guides learners to see that $x = 0$ is simply the y-axis acting as a mirror wall, and $y = 0$ is the x-axis acting as a mirror floor, so the algebraic rules are not arbitrary but follow inevitably from the geometry already understood.

By the end of the lesson learners can predict where any polygon's image will land under reflection in either axis without constructing perpendiculars — they simply apply

$(x, y) \rightarrow (-x, y)$ or $(x, y) \rightarrow (x, -y)$. This quantitative power advances the driving question: the "invisible rule" governing where the image lands is now expressed as a coordinate transformation, and the reason the image is the same size but laterally flipped is encoded in the sign change on exactly one coordinate.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reflecting points in the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reflect points Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner receives a half-sheet showing a Cartesian plane with a single triangle ABC plotted in Quadrant I ($A = (2, 3)$, $B = (5, 3)$, $C = (5, 7)$). Without any instruction, learners individually write down: (1) the coordinates they predict the image triangle A'B'C' will have after reflection in the y-axis ($x = 0$), and (2) the coordinates they predict after reflection in the x-axis ($y = 0$). They record their reasoning in one sentence next to each prediction. After 3 minutes learners share predictions with a partner and note any disagreements on a sticky note to post on the Driving Question Board.	Distribute the half-sheet and say: 'You have already built images with a ruler and set-square. Now the mirror is the y-axis — the vertical line down the middle of your grid. Without drawing anything yet, predict the coordinates of the image. Write your prediction and your reason. You have 3 minutes — work alone.' [Allow 3 minutes of silent individual thinking. Circulate and note 2–3 interesting or divergent predictions without correcting them.] After 3 minutes say: 'Compare with your partner. If you disagree, write your disagreement on the sticky note provided.' [Allow 1 minute for partner talk.] Cold-call 3 pairs: 'Pair at the front — what did you predict for A' when you reflect in the y-axis, and why?' [WAIT TIME: 12 seconds after each question before accepting a response.] Do NOT confirm or deny predictions. Say: 'Hold those ideas — we are about to gather evidence.'	Predict–Record–Compare: Learners commit written predictions to paper before observing, activating prior knowledge of the distance-and-perpendicularity properties from Lesson 2. Recording forces articulation of reasoning, not just a guess, and the partner comparison surfaces misconceptions (common prediction: image at $(2, -3)$ instead of $(-2, 3)$ for y-axis reflection) that the Observe phase will resolve.	Teacher scans written predictions while circulating. Key indicator: does the learner apply 'same distance, opposite side' correctly to both coordinates or only to one? A learner who writes $A' = (-2, 3)$ for y-axis reflection shows correct geometric intuition; a learner who writes $(2, -3)$ is confusing the two axes. Teacher mentally flags these learners for targeted cold-calling during the Explain phase.
Observe Phase  VIDEO: Reflecting points in the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners work in pairs. Each pair receives a large-squared graph paper (or uses the ARES grid tool) showing a 'Nairobi School Grounds Grid Map' with four labelled buildings: the school gate at $(1, 1)$, the science lab at $(4, 1)$, the library at $(4, 5)$, and the staffroom at $(1, 5)$ — forming a rectangle GSLF. Learners are	Distribute the graph paper and say: 'This grid represents part of the grounds of a school in Nairobi — the y-axis is the school's main wall and the x-axis is the school fence. Plot the four buildings, then use everything you learnt in Lesson 2 — perpendiculars, equal distances — to reflect the rectangle in each axis. Fill in the	Data Table Pattern Recognition: By recording object and image coordinates in a structured three-column table — a classic science-and-engineering practice of analysing data — learners see the numerical pattern emerge across four data points simultaneously. The Nairobi school-grounds context anchors the abstract	Teacher checks tables for accuracy of all 8 image coordinates (4 per axis). Correct entries: reflection in $x=0$ gives $G'=(-1,1)$, $S'=(-4,1)$, $L'=(-4,5)$, $F'=(-1,5)$; reflection in $y=0$ gives $G''=(1,-1)$, $S''=(4,-1)$, $L''=(4,-5)$, $F''=(1,-5)$. Learners who swap sign changes (e.g., applying x-axis rule to y-axis reflection) need

 EXERCISE: Reflect points Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	asked to: (1) plot and label the rectangle GSLF; (2) reflect it in the y-axis ($x = 0$) and plot the image G'S'L'F'; (3) reflect the original in the x-axis ($y = 0$) and plot image G"S"L"F"; (4) complete a three-column coordinate table (Object Image in $x=0$ Image in $y=0$) for all four vertices. They then look for the pattern and write a generalisation in words before converting it to algebraic notation.	table as you go. You have 10 minutes.' [Circulate. After 5 minutes, pause the class for 30 seconds: 'Stop and look at your table so far. What is happening to the x-coordinate when you reflect in the y-axis? What is happening to the y-coordinate?' [WAIT TIME: 15 seconds.] Cold-call 2 pairs to share partial observations. Then say: 'Continue — finish the table and write your generalisation.' After 10 minutes total, bring the class together. Ask: 'Who can tell me the pattern in one sentence — what changes and what stays the same?' [WAIT TIME: 12 seconds.] Cold-call 3 individuals. Write their exact words on the board without editing, then ask: 'Can we write that using algebra? What would (x, y) become?'	coordinates in a familiar spatial layout, making the sign-change rule feel geometrically motivated rather than arbitrary.	immediate pair-level intervention. Teacher also checks that generalisations are written before algebra is attempted — a learner who jumps straight to $(-x, y)$ without a verbal statement has not yet consolidated the geometric meaning.
Explain Phase  VIDEO: Reflecting shapes: diagonal line of reflection <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners apply the two coordinate rules to a new challenge: a triangle PQR representing the matatu from the phenomenon ($P = (3, 2)$, $Q = (6, 2)$, $R = (6, 6)$) is reflected in the y-axis to model the view in the driver's side mirror (the mirror wall = y-axis), and then the same triangle is reflected in the x-axis to model the rear-view mirror view (mirror floor = x-axis). Learners write the image coordinates using the rules — no construction required — and then verify one image by checking perpendicularity and equal distance for vertex P only (using the Lesson 2 method). Finally, learners write a concise two-sentence explanation connecting the sign-change rule back to Aisha's bedroom mirror: why does her image raise the opposite hand?	Say: 'You now have the rules. Use them — no ruler, no perpendiculars — just the algebra — to write down the image of triangle PQR under reflection in $x = 0$ and then in $y = 0$.' [Allow 4 minutes.] After learners have written image coordinates, say: 'Now verify your answer for vertex P only using the perpendicular-and-equal-distance method from Lesson 2. This check tells us the rule is consistent with the geometry.' [Allow 3 minutes.] Cold-call 1 learner: 'Read me your image coordinates for the y-axis reflection.' [WAIT TIME: 10 seconds.] Write them on the board. Ask the class: 'Thumbs up if you agree, thumbs to the side if you are unsure.' Address any thumbs-to-the-side directly. Then ask: 'Here is the big question — look at your image of P under $x = 0$. The x-coordinate changed sign.	Rule–Verify–Connect: Learners first apply the algebraic rule (efficiency), then verify it geometrically (validity), then connect it to the anchoring phenomenon (meaning). This three-step sequence, adapted from the NGSS practice of using mathematics and computational thinking alongside constructing explanations, ensures the rule is not memorised in isolation but understood as an inevitable consequence of the geometry of mirrors.	Teacher looks for three things: (1) Correct image coordinates for PQR under both axes — $P'=(-3,2)$, $Q'=(-6,2)$, $R'=(-6,6)$ for $x=0$; $P''=(3,-2)$, $Q''=(6,-2)$, $R''=(6,-6)$ for $y=0$. (2) A valid perpendicularity check for vertex P showing PP' is horizontal (same y, opposite x) and midpoint of PP' lies on $x=0$. (3) The written explanation must explicitly name 'the x-coordinate changes sign' as the cause of the lateral flip. A vague answer ('the image is reversed') is flagged for follow-up.

		In Aisha's bedroom, she raises her right hand and the image raises what appears to be a left hand. Which coordinate is responsible for that flip, and why?' [WAIT TIME: 15 seconds.] Cold-call 2 learners. Conclude by writing on the board: 'Reflection in $x = 0$: $(x, y) \rightarrow (-x, y)$ — the horizontal distance flips, the vertical stays. Reflection in $y = 0$: $(x, y) \rightarrow (x, -y)$ — the vertical distance flips, the horizontal stays.'		
Driving Question Board (DQB) Creation  VIDEO: Determining reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners retrieve their sticky notes from the Predict phase. Each pair now writes two new sticky notes: one labelled 'NOW I KNOW' stating the coordinate rule they have confirmed, and one labelled 'STILL WONDERING' raising a new question sparked by today's work (e.g., 'What happens if the mirror line is $y = x$ instead of an axis?' or 'Can you reflect in a line that is not through the origin?'). Pairs post both notes on the class DQB under the columns 'Evidence Gathered — Lesson 4' and 'New Questions'. The class does a 2-minute silent gallery walk of the 'STILL WONDERING' notes.	Point to the DQB and say: 'Our driving question asks where exactly the copy in the mirror lives. Today we found a coordinate address for it. Write on your green sticky note exactly what you now know about that address — use the algebraic rule. On your orange sticky note, write one question that today's lesson opened up that we have not yet answered.' [Allow 2 minutes for writing.] As pairs post notes, read two 'STILL WONDERING' notes aloud without attributing them: 'Someone is asking — what if the mirror line is diagonal? Another person asks — does the rule change if the shape is in Quadrant III? These are excellent questions. Our storyline will visit them.' Point to the DQB column for Lesson 5 and say: 'We will come back to those in our next lesson.'	Know–Wonder Board Progression: The DQB is a cumulative public record of the class's growing understanding across all 8 lessons. By explicitly distinguishing 'NOW I KNOW' (evidence-based claims) from 'STILL WONDERING' (productive questions), learners practise the scientific practice of distinguishing what is established from what remains open — directly mirroring how mathematicians build theory incrementally.	Teacher reads all 'NOW I KNOW' sticky notes during the gallery walk. A correct entry names the rule symbolically, e.g., $(x, y) \rightarrow (-x, y)$ for $x = 0$. An incomplete entry such as 'the coordinates change' indicates the learner has not consolidated the rule and needs a follow-up question in the next lesson's warm-up. Teacher photographs the DQB state for the lesson record.
Model Building Phase  VIDEO: Determining reflections <i>Related topic: not</i>	Learners return to their cumulative unit model — a labelled diagram of Aisha's bedroom scenario that has been updated in each lesson. Today they add a Cartesian overlay: they sketch a small coordinate grid on their model,	Say: 'Update your model. Add the coordinate grid, place Aisha and her image, write the rule, and annotate why the hand appears to flip. You have 4 minutes.' [Circulate. Look specifically for whether learners place the mirror	Cumulative Model Revision: The model is not redrawn each lesson but annotated and extended, making conceptual growth visible. Adding a coordinate overlay to the existing physical diagram makes explicit the connection between	Teacher checks three elements in each learner's model: (1) Mirror line correctly drawn as the y-axis ($x = 0$), not as a freehand vertical line. (2) Aisha's image coordinates correctly stated as $(-3, 0)$. (3) The annotation explicitly links the sign

<i>an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Reflections in a Plane Mirror (PLIX) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	place Aisha at point (3, 0) and the mirror wall along the y-axis ($x = 0$), then write her image's coordinates as $(-3, 0)$ and annotate: 'Rule: $(x, y) \rightarrow (-x, y)$. The mirror at $x = 0$ flips the x-coordinate, which is why Aisha's image raises the opposite hand — the left-right direction is the x-direction.' Learners also add an arrow labelled '$2 \times 3 = 6$ m total distance' to connect to the equal-distance property from Lesson 2. Finally, learners write one sentence at the bottom of their model updating their best current answer to the driving question.	on $x = 0$ rather than between two grid lines, and whether the annotation explains the sign change geometrically.] After 4 minutes, cold-call 2 learners to read their driving-question update sentence aloud. [WAIT TIME: 12 seconds before accepting.] Affirm precision: 'I hear you saying the image lives at $(-3, 0)$ because the rule flips the sign of x. That is the most precise answer we have given to our driving question so far — the copy lives at a coordinate address that is the mirror image of the object's address.' Say: 'In Lesson 5 we will move the mirror to the line $y = x$ and see if our model needs updating again.'	the geometric construction method (Lessons 2–3) and the algebraic rule (Lesson 4), embodying the NGSS practice of developing and using models as explanatory tools that evolve with evidence.	change in x to the lateral flip observed in the phenomenon. A model that shows correct coordinates but no explanatory annotation is rated 'Approaches Expectations' and earns a written prompt: 'Why does the x-coordinate change sign? What does that mean for Aisha's hand?'
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D. TEACHER REFLECTION

1. During the Observe phase, did learners complete the coordinate table with sufficient accuracy to see the sign-change pattern independently, or did most pairs need a hint? If hints were needed, consider pre-teaching the table structure as a warm-up in the next lesson.
2. Were learners able to articulate the difference between reflection in $x = 0$ (sign of x changes) and reflection in $y = 0$ (sign of y changes), or did a significant number confuse the two? Prepare a targeted 5-minute warm-up for Lesson 5 if more than one-third of the class conflated them.
3. Did the Nairobi school-grounds context make the coordinate work feel grounded and meaningful, or did learners treat it as a superficial label? Note any moments where a learner spontaneously connected the grid to a real spatial layout — these indicate the context is working.
4. How precise were learners' 'NOW I KNOW' sticky notes? Did most write the algebraic rule $(x, y) \rightarrow (-x, y)$, or only a verbal description? The level of symbolic precision on the DQB tells you whether the class is ready for the more abstract $y = x$ mirror line in Lesson 5.
5. Did any learner's driving-question update sentence advance beyond 'the copy lives at $(-x, y)$ ' to include an explanation of why (because the mirror is perpendicular to the x -direction and equidistant)? If so, use that sentence as the opening provocation for Lesson 5.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When triangle and rectangle vertices in Quadrant I were reflected in the y -axis ($x = 0$), only the x -coordinate changed sign while y stayed the same; when reflected in the x -axis ($y = 0$), only the y -coordinate changed sign while x stayed the same — confirmed across all eight vertices of the Nairobi school-grounds rectangle.
What did I learn?	Reflection in the y -axis follows the rule $(x, y) \rightarrow (-x, y)$; reflection in the x -axis follows the rule $(x, y) \rightarrow (x, -y)$. In both cases the mirror line is the perpendicular bisector of the segment joining each object point to its image, consistent with the three properties established in Lesson 2.
How does this explain the	The coordinate rules explain precisely why Aisha's image appears laterally reversed: standing at $x = 3$ in front of her mirror wall

phenomenon?	(the y-axis, $x = 0$), her image is located at $x = -3$, so the left–right (x) direction is flipped while the up–down (y) direction is preserved — which is exactly why the image raises the opposite hand. The 'invisible rule' governing where the image lands is now a coordinate transformation.
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LESSON 5 (40 minutes): Reflection on the Cartesian Plane II — The $y = x$ Mirror Line

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners investigate reflection in the line $y = x$ on the Cartesian plane, deduce the coordinate transformation rule $(x, y) \rightarrow (y, x)$, and consolidate all three Cartesian mirror lines ($x = 0$, $y = 0$, $y = x$) into a unified reference framework.
Knowledge	Learners will know that reflection in the line $y = x$ maps every point (x, y) to the image point (y, x) ; that the mirror line $y = x$ is the perpendicular bisector of the segment joining each object point to its image; and that the three Cartesian mirror lines $x = 0$, $y = 0$, and $y = x$ each produce a distinct, rule-governed coordinate transformation.
Skills	Learners will be able to: plot object points and their images under reflection in $y = x$; state and apply the rule $(x, y) \rightarrow (y, x)$; draw the mirror line $y = x$; verify reflection properties (equal distance, perpendicularity) for the new mirror line; and select and apply the correct transformation rule when given a mixed set of Cartesian reflection problems.
Attitudes	Learners will appreciate that every mirror line — whether physical (the bedroom mirror, the matatu rear-view mirror) or abstract (a line on the Cartesian plane) — imposes a precise, predictable rule on where an image lands, building confidence in using mathematics to explain everyday observations.
Key Inquiry Question	How is reflection important in day-to-day activities?
Purpose in Storyline	In Lessons 3 and 4 learners discovered that the y -axis ($x = 0$) and x -axis ($y = 0$) mirror lines swap the sign of one coordinate, just as a physical mirror swaps left and right or near and far. Lesson 5 introduces a diagonal mirror — the line $y = x$ — and shows that this line swaps the two coordinates entirely, $(x, y) \rightarrow (y, x)$. This mirrors the anchoring phenomenon: when Aisha tilts the full-length mirror on her bedroom wall at 45° , her reflection is transformed in a completely new way. By comparing all three mirror lines in a summary table, learners build a complete geometric 'rule book' for plane reflection and move significantly closer to answering the driving question about where exactly the mirror-image copy lives.
Safety Notes	No physical hazards. Remind learners to handle rulers and geometrical set instruments with care. If graph paper is distributed as printed sheets, ensure scissors or sharp objects are not needed for this lesson.

B. LESSON OVERVIEW

In this 40-minute lesson learners first predict what happens to coordinates when a point is reflected in the diagonal line $y = x$, using their knowledge of the bedroom mirror tilted at 45° . They then gather evidence by plotting three object points and their reflections on squared/graph paper (or a shared Cartesian grid on the board), measuring distances, and tabulating results. Through guided sensemaking they induce the rule $(x, y) \rightarrow (y, x)$, verify it geometrically, and consolidate it alongside the $x = 0$ and $y = 0$ rules in a three-row summary table. A mixed-problem practice set and a DQB update close the lesson. Throughout, the anchoring phenomenon — Aisha's mirror, the matatu side mirrors, and the letter 'A' experiment — is explicitly revisited to ground every abstract rule in observable reality.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the anchoring	1. Display the anchoring	Elicit-and-record prediction: By	Observable indicator: Every

 VIDEO: Reflecting shapes: diagonal line of reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Images in a Plane Mirror (Flexbook) Source: CK-12  Search ARES for similar readings	phenomenon: Aisha's mathematics teacher held a bold letter 'A' beside a plane mirror and the class saw a laterally reversed image. The teacher now poses a new scenario: 'Imagine Aisha tilts that same mirror so it sits exactly along the diagonal line $y = x$ on a Cartesian grid — at 45° to both axes. She places a point at (2, 5). Where do you predict the image will land?' Learners write their individual prediction (coordinate pair) in their exercise books, then share with a shoulder partner. Selected pairs are cold-called to share predictions with the class (teacher cold-calls 4 pairs). Predictions are displayed on a class 'Prediction Wall' (a section of the board left blank for this purpose) without judgment.	phenomenon recap (30 seconds): 'In our last two lessons we found rules for the x-axis and y-axis mirror lines. Today we add a new mirror line — the diagonal $y = x$ — just like tilting Aisha's bedroom mirror at 45°.' 2. Draw a clear Cartesian grid on the board with the line $y = x$ drawn in a contrasting colour. Mark point A(2, 5) visibly. 3. Pose the prediction prompt verbatim: 'Write down the coordinates you predict for the image of point A(2, 5) when it is reflected in the line $y = x$. You have 60 seconds — work silently and independently.' 4. Allow 60 seconds silent individual thinking. 5. 'Now turn to your neighbour and compare your predictions. Agree on ONE coordinate pair between you. 30 seconds.' 6. WAIT TIME: After calling 'ready', wait 10 seconds before cold-calling. 7. Cold-call exactly 4 pairs by name: 'Kariuki and Njeri — what did you predict?', 'Otieno and Akinyi?', 'Wanjiru and Mwangi?', 'Fatuma and Hassan?' 8. Record all four predictions on the Prediction Wall without commenting on correctness. 9. Ask: 'What pattern, if any, do you notice in your classmates' predictions?' Wait 10 seconds, take 2 volunteer responses.	committing to a coordinate pair before instruction, every learner activates prior knowledge of the $x = 0$ and $y = 0$ rules and forms a personal hypothesis. Recording predictions publicly creates cognitive dissonance that the Observe phase will resolve, deepening retention of the correct rule.	learner has written a coordinate pair in their book within 60 seconds. Teacher scans the room during the silent minute to check participation. The spread of predictions on the Prediction Wall (e.g., whether learners offer (5, 2), (-2, -5), (-5, -2), or random guesses) reveals existing misconceptions about diagonal reflections that the teacher notes for targeted follow-up during the Explain phase.
Observe Phase  VIDEO: Reflecting points in the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners receive (or open to) a pre-drawn Cartesian grid (x and y from -6 to 6) with the line $y = x$ already drawn. Working in pairs, they complete a structured evidence-gathering table: they plot three object points — A(2, 5), B(-1, 3), C(4, -2) — then use a ruler to draw the perpendicular from each point to the line $y = x$, extend it an equal distance on the	1. Distribute grids (or direct learners to draw on graph paper). Confirm the line $y = x$ is visible: 'The line $y = x$ passes through the origin at 45°. Check: does your line pass through (1,1), (2,2), (3,3)? Give me a thumbs-up when you can see that.' 2. Model the construction for point A(2, 5) live on the board: draw the perpendicular from A to the line $y =$	Structured data-collection table: Requiring learners to record both the coordinate evidence AND the distance measurements simultaneously connects the algebraic rule to its geometric justification. The distance measurements directly mirror the anchoring phenomenon observation that 'the image is the same distance behind the mirror	Observable indicator: Learner evidence tables should show A'(5, 2), B'(3, -1), C'(-2, 4) with equal perpendicular distances (to within ± 0.2 cm measurement tolerance). Teacher checks at least 6 pairs' tables during circulation. Pairs whose image coordinates are not the correct swap are flagged — the teacher asks 'Can you count grid squares from your point to the

 EXERCISE: Reflect points Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	other side, and mark the image points A', B', C'. They measure and record: (i) the object coordinates, (ii) the image coordinates they find by construction, (iii) the perpendicular distance from object to mirror line, and (iv) the perpendicular distance from image to mirror line. They record all findings in a four-column evidence table in their books.	x (remind learners how to use a set-square for perpendicularity), measure the distance, mark the image A' on the other side at the same distance. Narrate aloud: 'I am measuring exactly 2.1 cm from A to the mirror line; I mark 2.1 cm on the other side — this is A'.' 3. Say verbatim: 'Now you do points B and C with your partner. Use your ruler carefully. You have 8 minutes.' 4. Circulate; crouch to pair level; ask probing questions: 'How do you know your perpendicular is truly at 90° to $y = x$?', 'Is the image closer to or further from the line than the object?' 5. After 8 minutes: 'Pause. Read off the coordinates of A', B', C' from your grid and fill in the evidence table. You have 2 minutes.' 6. WAIT TIME before whole-class share: 10 seconds of silence after saying 'Pens down.' 7. Cold-call 3 individual learners (not pairs this time) to read their image coordinates aloud: 'Chebet, what are the coordinates of A?', 'Mwenda, what did you get for B?', 'Adhiambo, and C?'. Record responses on the board in a class evidence table next to the objects.	as the object is in front' — now verified quantitatively on the Cartesian grid.	mirror line and back?' to redirect without giving the answer.
Explain Phase  VIDEO: Reflecting shapes: diagonal line of reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Geometry Software for Reflections (Flexbook)	The teacher guides the class to extract the pattern from their evidence table. Learners compare object and image coordinate pairs, state the rule in words, write it in algebraic notation, and verify it against their Prediction Wall entries. They then complete a three-row Reflection Rules Summary Table comparing all three mirror lines studied. Finally, they apply the three rules to a set of four mixed practice problems, self-marking against answers revealed by the teacher.	1. Point to the class evidence table on the board: $A(2,5) \rightarrow A'(5,2)$; $B(-1,3) \rightarrow B'(3,-1)$; $C(4,-2) \rightarrow C'(-2,4)$. Ask verbatim: 'Look carefully at the object coordinates and the image coordinates. What do you notice? Write one sentence in your book. 30 seconds.' WAIT 10 seconds after pens go down before taking responses. Cold-call 4 learners. Steer toward: 'The x and y coordinates swap places.' 2. Ask: 'How do we write this as a mathematical rule? Try writing it	Pattern-induction followed by rule consolidation in a comparison table: Learners first discover the $y = x$ rule from their own data (inductive reasoning), then slot it into a growing framework alongside $x = 0$ and $y = 0$. The comparison table is a Knowledge Organiser — a single visual that encodes all three rules and allows learners to self-quiz. The mixed practice immediately transfers the rules to varied contexts, embedding procedural fluency alongside conceptual	Observable indicators: (i) All learners can write $(x,y) \rightarrow (y,x)$ correctly in their books — teacher checks 8 books during the table-filling activity. (ii) In mixed practice, the teacher notes which problem type produces the most errors (commonly Q(b) or Q(c) — sign errors on axes) and names the misconception aloud: 'I notice some learners changed BOTH coordinates for the x-axis reflection — remember, only the x-coordinate changes sign for $x = 0$.' (iii) The show-of-hands score on

Source: CK-12 Search ARES for similar readings		using $(x, y) \rightarrow (?, ?)$.' Wait 15 seconds. Cold-call 2 learners. Confirm and write prominently on the board: RULE: Reflection in $y = x$ maps $(x, y) \rightarrow (y, x)$. 3. Return to the Prediction Wall: 'Which pairs predicted (5, 2) for A(2,5)? Stand up.' Celebrate correct predictors. Ask one incorrect predictor: 'What did you predict, and why do you think the rule is (y, x) rather than your answer?' (No judgment; frame as 'our prediction helped us notice the pattern.') 4. Distribute or display the three-row Reflection Rules Summary Table: headers — Mirror Line Rule Example. Learners fill in all three rows: $x=0$: $(x,y) \rightarrow (-x,y)$; $y=0$: $(x,y) \rightarrow (x,-y)$; $y=x$: $(x,y) \rightarrow (y,x)$. Give 3 minutes. 5. Phenomenon link — say verbatim: 'Think back to Aisha's matatu ride. The side mirror showed vehicles behind her. If the road is like the x-axis and the mirror is the y-axis, the car at position (3, 1) appears at (-3, 1). But if Aisha tilted that mirror to 45°, the car would appear at (1, 3). Different mirror angle — different rule — but always precise and predictable. That is what mathematics gives us.' 6. Mixed practice: display 4 problems (2 minutes each, individual silent work): (a) Reflect P(3,1) in $y=x$; (b) Reflect Q(-2,4) in $x=0$; (c) Reflect R(0,5) in $y=0$; (d) Reflect S(-3,-1) in $y=x$. After 8 minutes, reveal answers; learners self-mark. 7. Ask: 'Raise your hand if you got all four correct.' Then: 'Raise your hand if you got three.' Use this as a live participation check.	understanding.	4/4 versus 3/4 gives a rapid class-level accuracy read.
Driving Question Board (DQB)	Each learner writes one new understanding on a green sticky note and one remaining question	1. Hand out one green and one yellow sticky note per learner (or instruct learners to tear a page into	Metacognitive DQB curation: The two-colour sticky-note protocol separates consolidated knowledge	Observable indicator: Green notes should include at least one of: the rule $(x,y) \rightarrow (y,x)$, a reference to

Creation  VIDEO: Reflecting points in the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Images in a Plane Mirror (Flexbook) Source: CK-12  Search ARES for similar readings	on a yellow sticky note, then physically adds them to the class DQB. The teacher reads selected notes aloud and explicitly links the lesson's findings back to the driving question: 'Why does a mirror show a perfect copy of the world — and where exactly does that copy live?'	two coloured-pencil-outlined boxes if physical notes are unavailable). 2. Say verbatim: 'On the GREEN note write: ONE thing you now know about where the image lives — be specific, use coordinates or the rule. On the YELLOW note write: ONE question you still have about reflections or mirror lines.' Allow 2 minutes silent writing. 3. Learners bring notes to the DQB (a section of the wall or a large piece of manila paper). Organise notes into two columns: WHAT WE NOW KNOW WHAT WE STILL WONDER. 4. Teacher reads 3 green notes aloud, e.g., 'Waweru writes: The image in $y=x$ is found by swapping the coordinates — (x,y) goes to (y,x).' Then reads 2 yellow notes, e.g., 'Akinyi asks: What happens if the mirror line is $y = 2x$ instead of $y = x$?' 5. Explicitly connect to driving question — say verbatim: 'Our driving question asks WHERE EXACTLY the copy lives. After five lessons, we now have three precise mathematical addresses for images in Cartesian mirrors — and we have verified them with rulers and coordinates. Akinyi's question — $y = 2x$ — is a preview of what more advanced mathematics can answer. Keep that question on the board.' 6. Point to the DQB and note the growing cluster of 'WHAT WE NOW KNOW' notes: 'Compare this column to where it was in Lesson 1. We are building a complete answer to our driving question, one mirror line at a time.'	(green) from productive uncertainty (yellow), making the boundary of understanding visible to both teacher and learner. Publicly reading selected notes validates diverse contributions and models scientific discourse — the practice of distinguishing evidence-based claims from open questions, directly paralleling the NGSS practice of 'Obtaining, Evaluating, and Communicating Information.'	equal distances, or a mention of the mirror line $y=x$. A green note that only says 'reflection swaps things' signals surface-level understanding — the teacher notes that learner for follow-up in Lesson 6. Yellow notes that reference new mirror lines (e.g., $y=-x$, $y=2$) indicate readiness for extension and inform Lesson 6 planning.
Model Building Phase  VIDEO: Determining	Learners retrieve the cumulative geometric model they have been building since Lesson 1 (a Cartesian plane diagram showing object, image, and mirror line).	1. Say: 'Take out your cumulative reflection model. Today we add the third and most powerful mirror line we have studied.' Allow 30 seconds for learners to locate their	Cumulative annotated model revision: Each lesson learners physically add one new element to a single evolving diagram. This models the scientific practice of	Observable indicator: Teacher checks that the line $y=x$ passes through $(0,0)$, $(1,1)$, $(2,2)$ — a common error is drawing it through $(0,1)$. The example point A' must

reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Reflect points Source: Khan Academy (English - US curriculum) Search ARES for similar readings	They add the $y = x$ mirror line in a new colour and annotate it with: (i) the algebraic rule $(x,y) \rightarrow (y,x)$, (ii) a plotted example showing $A(2,5)$ and $A'(5,2)$ with the perpendicular bisector drawn, and (iii) a one-sentence explanation of how this new mirror line changes the image compared to the x-axis and y-axis mirror lines. Learners who finish early are invited to also sketch what happens to a small triangle (with vertices at $(1,2)$, $(3,2)$, $(3,4)$) reflected in $y=x$, labelling all image vertices.	models. 2. Demonstrate on the board: draw the line $y=x$ in orange (or any new colour) on the existing model, plot $A(2,5)$, draw the perpendicular to $y=x$, mark $A'(5,2)$, and write the rule beside the mirror line. Talk aloud: 'I am adding $y=x$ in orange so I can immediately see all three mirror lines at a glance — blue for $x=0$, red for $y=0$, orange for $y=x$.' 3. Learners work independently for 5 minutes to update their own model. Teacher circulates and checks: (a) Is the line $y=x$ correctly drawn at 45° through the origin? (b) Is the example point and its image correctly placed? (c) Is the rule written accurately? 4. For the one-sentence comparison, prompt with: 'How is the $y=x$ image different from the $x=0$ image of the same point? Think about $A(2,5)$: where does it go for each mirror line?' WAIT 10 seconds. Take 2 volunteer responses before learners write. 5. With 2 minutes remaining, close by saying verbatim: 'Your model now holds three reflection rules. In our next lesson we will use this model to find the mirror line when we are only given the object and its image — working backwards, just like a detective. Keep your model safe.' 6. Extension task (stated aloud for fast finishers): 'Reflect the triangle with vertices $K(1,2)$, $L(3,2)$, $M(3,4)$ in the line $y=x$. Label all image vertices and check that the image triangle is congruent to the original.'	'Developing and Using Models' — the model is never 'finished' but is refined with each new piece of evidence. The colour-coding (a different colour for each mirror line) creates a visual knowledge organiser that makes the three rules simultaneously comparable and aids long-term retrieval. The one-sentence comparison forces learners to articulate the conceptual difference between rules, not just apply them procedurally.	be at $(5,2)$, not $(-2,-5)$ or another confusion. The one-sentence comparison should explicitly mention that $x=0$ negates the x-coordinate while $y=x$ swaps both coordinates — a surface-level answer ('they are all reflections') prompts the teacher to ask: 'Show me numerically, using $A(2,5)$ as your example, what is different.' The extension triangle task is collected from fast finishers and used as a stretch indicator for Lesson 6 grouping.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did learners successfully induce the rule $(x,y) \rightarrow (y,x)$ from their own plotted data, or did most need the teacher to

tell them the rule directly? If the latter, consider providing a more scaffolded evidence table in Lesson 6 that explicitly asks 'What do you notice about the x-value of the object versus the y-value of the image?' (2) Which of the three Cartesian mirror lines caused the most confusion in the mixed practice? Plan a 5-minute warm-up at the start of Lesson 6 targeting that specific rule. (3) Were all learners able to draw the line $y=x$ correctly at 45° through the origin? If not, a brief graphing-skills mini-lesson may be needed before Lesson 6's 'finding the mirror line' work. (4) Did the DQB yellow notes reveal any profound questions (e.g., about non-standard mirror lines such as $y=-x$) that could enrich Lesson 6 or be set as a bonus investigation for high-readiness learners? (5) Was 40 minutes sufficient for the model-building update, or did the Explain phase's mixed practice overrun? Adjust timing by reducing mixed-practice problems from 4 to 3 if needed.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners plotted three object points (A(2,5), B(-1,3), C(4,-2)) on a Cartesian grid and used a ruler to construct their reflections in the line $y = x$, recording both the image coordinates and the perpendicular distances from each point to the mirror line.
What did I learn?	Learners induced and stated the rule that reflection in the line $y = x$ maps every point (x, y) to the image point (y, x) — the two coordinates swap; they verified that the mirror line is the perpendicular bisector of the segment joining object to image; and they completed a three-row summary table comparing the transformation rules for all three Cartesian mirror lines studied ($x = 0$, $y = 0$, $y = x$).
How does this explain the phenomenon?	Learners explained that a diagonal mirror at 45° (the line $y = x$) swaps the two coordinates of a point, in contrast to the y-axis mirror (which negates the x-coordinate) and the x-axis mirror (which negates the y-coordinate); they connected this to the anchoring phenomenon by articulating that any mirror line — whether Aisha's physical bedroom mirror, the matatu side mirror, or a line on the Cartesian plane — imposes a precise, predictable mathematical rule that determines exactly where the image copy lives.

LESSON 6 (80 minutes): Finding the Mirror Line: Reverse Reflection on Plane Surfaces and the Cartesian Plane

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners work backwards from a known object and its image to construct or identify the mirror line as the perpendicular bisector of the segment joining each pair of corresponding points, deepening their understanding of all reflection properties simultaneously.
Knowledge	The learner knows that: (i) the mirror line is the perpendicular bisector of the segment joining any object point to its corresponding image point; (ii) this relationship holds on both a plain surface and the Cartesian plane; (iii) the mirror line can be uniquely determined from any single pair of corresponding points.
Skills	The learner is able to: (i) construct the perpendicular bisector of a segment joining corresponding object–image points to locate the mirror line on a plain surface; (ii) identify or draw the mirror line on a Cartesian plane given an object and its image; (iii) verify the mirror line by checking that it is equidistant from, and perpendicular to, the line joining every corresponding pair of points.
Attitudes	The learner appreciates that the geometric rule governing mirror reflection is precise, reversible, and directly applicable to real-life engineering decisions such as the positioning of road-safety junction mirrors.
Key Inquiry Question	How is reflection important in day-to-day activities?
Purpose in Storyline	In Lessons 1–5 learners established reflection properties and drew images given objects and mirror lines. Lesson 6 reverses the task: given the object and image, learners must reconstruct the invisible mirror line — the same challenge faced by traffic engineers who must determine exactly where to mount a junction mirror so that a driver (object) can see a specific road hazard (image). This reverse-engineering lens reframes the driving question: the 'perfect copy' in the mirror is governed by a precise geometric rule, and knowing that rule lets us place mirrors deliberately, not just observe them passively.
Safety Notes	Ensure all geometrical instruments (compasses, rulers) are handled carefully. No sharp-instrument injuries during construction activities. Learners working in pairs should share instruments without snatching.

B. LESSON OVERVIEW

Learners begin by revisiting the Nairobi bedroom-mirror and matatu-mirror phenomenon and are challenged with a new puzzle: if you can see the reflection but the mirror has been removed, can you reconstruct exactly where it stood? Through a structured Predict–Observe–Explain–DQB–Model sequence, learners use compass-and-ruler construction (and coordinate geometry on the Cartesian plane) to find mirror lines as perpendicular bisectors of object–image segments. The road-safety junction-mirror scenario grounds the abstract construction in an authentic Kenyan engineering context. By the end of the lesson, learners can both construct the mirror line on plain paper and identify it algebraically on the Cartesian plane, completing the full two-way mastery of reflection.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Geometric	Each learner receives a plain A4 sheet on which two congruent triangles (labelled ABC and A'B'C')	Distribute pre-drawn sheets before the lesson begins to save time. Project the same diagram on the	Prediction-with-evidence prompting. By requiring learners to articulate the visual clues that	Observable indicator: learner sketches show the predicted mirror line passing roughly midway

constructions: perpendicular bisector Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Determine reflections Source: Khan Academy (English - US curriculum) Search ARES for similar readings	have been pre-drawn in identical orientation but laterally flipped, with no mirror line shown. Learners study the sheet silently for 90 seconds, then write individual responses to three prompts in their exercise books: (1) 'Where do you think the mirror line is — sketch it freehand.' (2) 'What clues in the diagram tell you where the mirror must be?' (3) 'How might a traffic engineer use this same reasoning to position a junction mirror on the Nairobi–Thika highway?' Learners then share predictions with a shoulder partner for 2 minutes before the teacher takes three cold-call responses.	board using a TDD or drawn on the chalkboard. Read the three prompts aloud slowly, then say: 'You have 90 seconds of silent thinking — no talking, no pencils yet. Just look.' After silent time: 'Now write your three responses. You have 3 minutes.' After writing: 'Turn to your neighbour. Share your sketch and your clues. 2 minutes.' [WAIT 2 minutes.] Cold-call: 'Amina — where did you sketch the mirror line and why?' [WAIT 10 seconds.] 'Brian — did your clue agree with Amina's or differ? Tell us one thing that was different.' [WAIT 10 seconds.] 'Zawadi — what did your group say about the traffic engineer question?' [WAIT 12 seconds.] Record three different predicted mirror-line positions on the board without correcting them. Say: 'We will return to these predictions at the end of the lesson to see who was closest — and more importantly, why.'	locate the mirror line before any instruction, this strategy activates prior knowledge from Lessons 1–5 (distance-preservation, perpendicularity) and creates cognitive dissonance when predictions diverge, motivating the observational phase.	between the two triangles (acceptable range: within the correct half of the page). Listen for use of vocabulary from previous lessons — 'equidistant,' 'perpendicular,' 'midpoint' — as evidence of prior learning transfer. Flag learners whose sketches place the mirror line outside the figure entirely for targeted support in Phase 2.
Observe Phase VIDEO: Determining reflections Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Reflect points Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Activity A — Plain-surface construction (30 min, pairs): Each pair receives a fresh worksheet showing three separate object–image pairs on plain paper: (i) a single point P and its image P'; (ii) a line segment MN and its image M'N'; (iii) a quadrilateral KLMN and its image K'L'M'N'. Using a ruler and compass, learners: (a) join P to P' and construct the perpendicular bisector — this IS the mirror line; (b) repeat for two pairs of corresponding vertices on MN/M'N' and verify that both perpendicular bisectors are the same line; (c) repeat for two pairs of corresponding vertices on the quadrilateral and check all bisectors coincide. Learners	Before Activity A, model ONE construction step on the board: draw two points X and X', join them, then say: 'Watch carefully — I am going to open my compass to MORE than half of XX'. Set it on X. Draw an arc above and below the line. Now same radius, set it on X'. Two more arcs. Where they cross — THAT is the perpendicular bisector. Why does this give us the mirror line? I want you to figure that out.' [WAIT 15 seconds for silent thinking.] Circulate during pair work. At the 10-minute mark, pause the class: 'Hold up your worksheets — I want to see the construction arcs. If I cannot see arcs, you are measuring instead of constructing — stop and try again.'	Worked-example fading with reverse task structure. The teacher provides one full construction model, then learners complete progressively more complex cases independently. The deliberate reversal of the standard task (finding the image FROM the mirror line) to the inverse task (finding the mirror line FROM the image) forces learners to interrogate the underlying geometric relationship rather than follow a procedure, building conceptual depth.	Observable indicators: (1) Construction arcs are visible and intersect correctly — absence of arcs indicates procedural shortcutting; address immediately. (2) Measured distances from object and image to the constructed mirror line are equal (tolerance: ± 1 mm) — unequal distances indicate a construction error. (3) On the Cartesian task, the mirror line equation $x = 0$ is identified correctly and the midpoint of PP' is calculated as (0, 1) — wrong midpoints suggest coordinate arithmetic errors. Use mini-whiteboards or scrap paper for learners to show midpoint calculations for a quick whole-class scan.

	record: the midpoint coordinates (measured with a ruler), the angle between PP' and the mirror line, and the distances from P and P' to the mirror line. Activity B — Cartesian plane task (15 min, individual): Learners are given a Cartesian grid with triangle PQR (P(2,1), Q(4,1), R(3,4)) and its image P'Q'R' (P'(-2,1), Q'(-4,1), R'(-3,4)). They: (a) plot both triangles; (b) join P to P' and find its midpoint; (c) find the equation of the perpendicular bisector; (d) repeat for Q–Q'; (e) confirm the mirror line equation is $x = 0$. Learners annotate their grid with the mirror line and its equation.	Targeted question to struggling pairs: 'You said the midpoint is here — measure from your midpoint to P, then to P'. Are they the same? What does that tell you?' [WAIT 10 seconds.] Before Activity B, say: 'Now we move to the Cartesian plane. Instead of a compass you will use algebra. The rule is the same — find the midpoint, find the perpendicular. Let's see if the mirror line equation surprises you.' After Activity B, cold-call three learners: 'Kamau — what equation did you get for the mirror line?' [WAIT 10 sec.] 'Fatuma — how did you check it was correct?' [WAIT 10 sec.] 'Otieno — does this mirror line match any line we have worked with before in this unit?' [WAIT 12 sec.] Connect to phenomenon: 'This is exactly what a traffic engineer does at a junction on Waiyaki Way — she knows where the driver sits and where the hazard is, so she solves for the mirror position mathematically.'		
Explain Phase  VIDEO: Reflecting points in the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Whole-class sensemaking discussion (10 min): The teacher projects (or draws) a summary diagram showing an object point A, its image A', the segment AA', its midpoint M, and the perpendicular bisector through M labelled as the mirror line ℓ. Learners are asked to articulate in their own words the complete logical chain: 'Why must the mirror line be EXACTLY the perpendicular bisector — not just any line through M?' Pairs discuss for 90 seconds, then three pairs share. Application task — Junction Mirror Problem (10 min, individual): Learners receive a printed map-style diagram (Kenya	Open the discussion: 'Before you tell me the answer, I want the REASON. Zawadi — if the mirror line just passed through the midpoint but was NOT perpendicular, what would go wrong with the reflection?' [WAIT 12 seconds.] Accept and probe responses. Say: 'The key word I want everyone to use in their explanation is EQUIDISTANT. The mirror line is the set of all points equidistant from the object and its image. Say that with me.' [Choral repetition.] 'Now write it in your book in your own words — 30 seconds.' For the Junction Mirror Problem, say: 'You are now the traffic engineer hired by Kenya	Reasoning-aloud with targeted vocabulary scaffolding. By requiring learners to articulate the logical chain using the term 'equidistant' and to write a real-world interpretation, the strategy consolidates procedural skill into conceptual understanding and links the abstract algebraic result to the anchoring phenomenon, making the mathematics meaningful rather than mechanical.	Observable indicators: (1) Learner explanations include both 'midpoint' and 'perpendicular' (or 'equidistant') — responses using only one term indicate partial understanding. (2) Junction Mirror equation is $x = -2$ (midpoint of (1,3) and (-5,3) is (-2,3); perpendicular bisector of a horizontal segment is a vertical line $x = -2$) — arithmetic errors in midpoint calculation are the most common failure mode; watch for learners averaging y-coordinates instead of x-coordinates. (3) Written real-world interpretations mention both the driver and the hazard, not just the equation — purely procedural responses

	road-junction context) showing: a driver's eye position D(1, 3) and a road hazard H(-5, 3) on a Cartesian grid. They must find the equation of the mirror line that would allow the driver to see the hazard in a junction mirror — i.e., find the perpendicular bisector of DH. Learners write a two-sentence 'real-world interpretation' beneath their working: 'The mirror should be placed along the line $x = __$ because ...'	Roads Board. Show your working as if you are writing a report. I need the equation of the mirror line AND one sentence explaining what it means physically.' Circulate and read two or three written interpretations aloud (anonymised) for class discussion. Cold-call: 'Brian — you wrote that the mirror should be at $x = -2$. Walk us through your midpoint calculation.' [WAIT 10 seconds.] 'Amina — does Brian's answer match the diagram? How do you know?' [WAIT 10 seconds.] Explicitly connect back to the phenomenon: 'Remember the matatu side mirror we discussed in Lesson 1? The engineer who designed the mirror mount solved exactly this kind of equation — they knew the driver's seat position and the danger zone position, and they found the mirror line mathematically.'		indicate the phenomenon connection has not been made.
Driving Question Board (DQB) Creation  VIDEO: Geometric constructions: perpendicular bisector Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reflect points Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each learner writes on a sticky note (or a designated section of their exercise book if sticky notes are unavailable) ONE new understanding and ONE remaining question prompted by today's lesson. They post or record these under two columns on the class DQB: 'What we now know' and 'What we still wonder.' The class then does a 3-minute silent gallery read of all posted notes. Three learners are cold-called to read aloud one 'still wonder' note that resonates with them and to suggest which future lesson might address it. The teacher annotates the DQB to show which questions will be answered in Lessons 7 and 8.	Say: 'At the start of this unit our driving question was: Why does a mirror show a perfect copy of the world — and where exactly does that copy live? On your sticky note, write: first — one thing you can now answer about that question that you could not answer in Lesson 1; second — one thing that still puzzles you.' [Allow 3 minutes writing time.] 'Stand up, post your note, then read three notes from classmates in silence. 3 minutes.' After gallery read, cold-call: 'Otieno — read one 'still wonder' note that you found interesting.' [WAIT 10 sec.] 'Fatuma — can you guess which lesson might answer that question?' [WAIT 10 sec.] 'Kamau — has today's lesson changed your answer to the driving question compared to what you wrote in Lesson 1? How?' [WAIT 12 sec.]	Metacognitive DQB curation. Requiring learners to distinguish between resolved and unresolved questions makes their growing understanding visible to themselves and to the teacher. The gallery-read ritual builds collective sense of progress and sustains curiosity for remaining lessons, maintaining the storyline coherence of the eight-lesson sequence.	Observable indicators: (1) 'What we now know' statements reference the perpendicular bisector relationship specifically — vague statements like 'mirrors work in a special way' indicate surface-level understanding. (2) 'Still wonder' questions are conceptually forward-looking (e.g., 'What happens when the mirror line is $y = x$?) rather than procedurally backward (e.g., 'How do I use a compass?') — procedural questions signal that Phase 2 consolidation was incomplete and require teacher follow-up. (3) Cold-call responses demonstrate ability to link today's learning to the original driving question.

		Point to the board: 'Our model of reflection is nearly complete. Lesson 7 will tackle reflections in the lines $y = 0$ and $y = x$ on the Cartesian plane, and Lesson 8 will bring everything together in a real road-safety design task.'		
Model Building Phase  VIDEO: Determining reflections Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners retrieve their cumulative 'Reflection Model' diagram that they have been building since Lesson 1 (a growing annotated diagram showing the mirror, the object, the image, and labelled properties). They add a new layer to the model: (a) draw a double-headed arrow between any object point and its image point, label the midpoint as lying ON the mirror line, and annotate 'mirror line = perpendicular bisector of every object-image segment'; (b) add a new box to their model's legend: 'REVERSE TASK: given object + image $\rightarrow$ construct mirror line using perpendicular bisector method'; (c) annotate the road-safety inset (a small junction-mirror sketch based on today's Junction Mirror Problem) with today's equation-based method, showing that the mirror line position can be calculated from known positions of driver and hazard. Pairs compare their updated models and identify any inconsistency with properties recorded in earlier lessons.	Say: 'Take out your Reflection Model from your portfolio. Today's lesson adds the final missing direction arrow — we can now go both ways: object $\rightarrow$ image AND image $\rightarrow$ object $\rightarrow$ mirror line. Add the new annotation now. You have 5 minutes.' [Circulate, checking that the perpendicular bisector annotation is correctly placed and that the road-safety inset is updated.] After 5 minutes: 'Swap models with your partner. Check: does their perpendicular bisector annotation match the properties we verified today? Does their road-safety inset use the correct equation method? Write one positive comment and one suggestion on a sticky note.' [Allow 3 minutes.] Cold-call final reflection: 'Amina — looking at your complete model from Lessons 1 to 6, what is the single most important geometric idea that explains everything the mirror does?' [WAIT 15 seconds.] 'Brian — how would you explain to the matatu driver why his side mirror shows vehicles that appear behind him but seem to be at the same distance as they really are?' [WAIT 12 seconds.] Close with the phenomenon link: 'Next lesson we will extend our model to mirror lines that are not just vertical or horizontal — and we will discover a new surprise about the line $y = x$.'	Cumulative model revision with peer audit. The physical act of annotating and then peer-auditing the cumulative model externalises understanding, makes conceptual growth tangible over the lesson sequence, and creates accountability for precision. Connecting the algebraic result to the road-safety inset already present in the model reinforces the phenomenon thread across all six lessons.	Observable indicators: (1) The perpendicular bisector annotation is correctly placed on the model with both the midpoint label and the right-angle symbol — missing either element indicates incomplete understanding of the two-part condition. (2) The road-safety inset shows an equation (e.g., $x = -2$) derived from the midpoint method, not just a freehand sketch — purely sketch-based updates indicate the algebraic method from Phase 3 has not been internalised. (3) Peer-audit sticky notes identify genuine mathematical inconsistencies rather than cosmetic comments — superficial comments ('nice diagram') suggest learners are not yet auditing for mathematical correctness; prompt with 'find one number or label to check.'

D. TEACHER REFLECTION

After the lesson, reflect on the following: (1) Did the reverse-task framing (finding the mirror line from the object and image) genuinely produce more conceptual discussion than forward tasks in Lessons 3–5, or did learners default to procedural compass use without articulating the perpendicular bisector reasoning? Note specific moments where the 'equidistant' vocabulary did or did not appear spontaneously. (2) In Activity B (Cartesian plane), did learners transfer the geometric construction reasoning into the algebraic midpoint-and-perpendicular method independently, or did they require significant scaffolding? Identify any learners who could do the plain-surface construction but could not set up the algebraic version — plan a bridging task for Lesson 7. (3) Was the Junction Mirror Problem (Kenya Roads Board scenario) sufficiently contextualised? Did learners write genuine real-world interpretations or merely re-state the equation? If interpretations were shallow, consider displaying a photograph of an actual Kenyan junction mirror (e.g., at a T-junction in Westlands, Nairobi) at the start of Lesson 7 to deepen the phenomenon connection. (4) Review the DQB 'still wonder' notes: are there patterns in remaining confusion (e.g., questions about non-vertical mirror lines, or about three-dimensional reflections) that should be pre-addressed in Lesson 7's predict phase? (5) Were lesson timings realistic? The construction activity is ambitious for 30 minutes — if pairs did not complete all three worksheet cases, prioritise cases (i) and (iii) in Lesson 7 warm-up review.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that when two congruent, laterally-flipped figures are drawn with no mirror line, the perpendicular bisectors of segments joining corresponding points all coincide at a single line — the mirror line — on both plain paper (via compass construction) and the Cartesian plane (via midpoint and perpendicular equation). They also observed that this reverse-engineering method can determine the exact placement of a junction mirror given the positions of a driver and a road hazard.
What did I learn?	Learners learned that the mirror line is the perpendicular bisector of the segment joining any object point to its corresponding image point; that this property holds universally on plain surfaces and the Cartesian plane; that the mirror line can be constructed geometrically using compass arcs or determined algebraically by finding the midpoint and the equation of the perpendicular line; and that this mathematical rule underpins the real-world positioning of road-safety mirrors by traffic engineers in Kenya.
How does this explain the phenomenon?	Learners explained why the mirror line must be both the midpoint locus AND perpendicular to each object–image segment — it is the set of all points equidistant from the object and its image, meaning any other line through the midpoint would produce a different, non-congruent image. They connected this reasoning to the anchoring phenomenon (the Nairobi bedroom mirror and matatu side mirrors) and to the junction-mirror engineering context, articulating that the 'invisible rule' governing where the reflection lives is the perpendicular bisector relationship, discoverable in both directions of the reflection task.

LESSON 7 (40 minutes): Real-Life Applications of Reflection — Group Investigation and Peer Presentations

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners consolidate and publicly communicate the geometric properties of reflection (Lessons 2–6) by investigating three Kenyan real-life applications — road-safety mirrors, Kanga/kitenge textile design, and architectural symmetry — and presenting their findings to the class.
Knowledge	– Reflection preserves congruence, perpendicularity from mirror line, and equal object/image distances — and these properties underpin every real-life mirror application. – Driving mirrors on Kenyan matatus and lorries use plane-mirror reflection to give drivers an undistorted, laterally reversed image of the road behind them. – Symmetry in Kanga/kitenge patterns and Kenyan architectural facades is created by reflecting a base motif or structural unit across one or more mirror lines. – The geometric rule (mirror line is the perpendicular bisector of the object–image segment) holds whether the context is a bedroom mirror, a road-safety mirror, a fabric print, or a building façade.
Skills	– Apply the three properties of reflection to analyse and annotate real-world examples from Kenyan contexts. – Prepare and deliver a short illustrated group presentation that links a chosen application to the formal geometric properties studied in Lessons 2–6. – Construct or sketch a labelled diagram showing object, mirror line, and image for a chosen real-life scenario. – Evaluate peers' presentations by identifying the specific reflection properties demonstrated.
Attitudes	– Appreciate that abstract geometric properties of reflection have concrete, life-saving and aesthetic value in Kenyan society. – Demonstrate responsibility and unity by contributing equitably to group research and presentation tasks. – Show respect for peers' perspectives by listening actively during presentations and providing constructive feedback.
Key Inquiry Question	How is reflection important in day-to-day activities?
Purpose in Storyline	Having built up the full geometric toolkit of reflection across Lessons 1–6 (lines of symmetry, three properties, plane-surface constructions, Cartesian rules, and finding mirror lines), learners now deploy that knowledge outward — examining how the same invisible geometric rule governs road safety on Kenyan highways, the beauty of Kanga textiles, and the grandeur of Kenyan buildings. This lesson answers the driving question directly: the 'perfect copy' that lives behind the mirror is not a mystery but a geometrically governed image, and its precise location and appearance serve real human purposes every single day.
Safety Notes	No sharp instruments or chemicals used. If printed/physical mirrors are available, remind learners to handle glass mirrors carefully and never to direct reflected sunlight toward anyone's eyes. Verbal and emotional safety: when discussing road accidents in the context of mirrors, frame discussion around safety solutions rather than graphic accident descriptions.

B. LESSON OVERVIEW

In this penultimate lesson of the reflection sub-strand sequence, learners move from mastery of geometric technique (Lessons 2–6) to real-world application and communication. The class is divided into three expert groups, each investigating a distinct Kenyan context in which the geometric properties of plane reflection are at work: Group A examines driving mirrors and road safety on Kenyan roads; Group B investigates bilateral and rotational symmetry in Kanga and kitenge textile design; Group C analyses architectural symmetry in landmark Kenyan buildings such as the Parliament Buildings in Nairobi and the ornate mosques of Mombasa's Old Town.

Each group prepares a short illustrated presentation — a labelled diagram or poster sketch, a worked example, and a verbal explanation — that explicitly links their application back to the three core properties of reflection established in Lesson 2 and the construction techniques practised in Lessons 3–6.

The lesson is structured around the ARES five-phase framework. The Predict phase activates prior learning by asking learners to brainstorm which everyday objects around them 'use reflection' and to predict which geometric property matters most in each case. The Observe phase is a structured 12-minute group research-and-sketch session using the application cards or resource pages provided on the ARES platform. The Explain phase consists of rapid 2-minute group presentations followed by class annotation of a shared diagram. The DQB update captures new answers and residual questions generated by seeing reflection in new contexts. The Model Building phase challenges each learner to add one new annotated real-life arrow to their cumulative reflection model, connecting phenomenon to geometry.

Throughout, the anchoring phenomenon — Aisha's bedroom mirror, the matatu rear-view mirror, and the letter 'A' in the classroom mirror — is kept visible. Each group is prompted to connect their application explicitly to one or more of these three phenomenon moments, ensuring the storyline closes coherently. By the end of the lesson every learner should be able to state confidently: 'The image appears exactly as far behind the mirror as the object is in front because the mirror line is the perpendicular bisector of the object–image segment — and this is true whether the mirror is on Aisha's bedroom wall, on a Nairobi matatu, printed onto a Kanga, or carved into the façade of a Mombasa mosque.'

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reflecting shapes: diagonal line of reflection <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12  Search ARES for similar readings	Learners open their exercise books to a fresh page and, individually and silently for 2 minutes, write or sketch answers to the prompt: 'List THREE places in your daily life — other than a bathroom mirror — where reflection is being used. For each one, predict which of the three properties of reflection (equal distances, perpendicular connector, congruent image) matters most, and why.' After individual think time, learners share with a shoulder partner for 1 minute. Selected pairs share with the class. The teacher records key predictions on the board under three emerging columns: Road/Transport, Art/Textiles, Buildings/Architecture.	Display the prompt on the board or ARES screen: 'Where does reflection appear in Kenyan daily life — beyond a bathroom mirror?' Say: 'You have two minutes to think and write on your own. No talking yet — I want YOUR ideas first.' [WAIT TIME: 2 minutes of silent individual writing.] After 2 minutes: 'Now share your list with the person next to you. You have 60 seconds.' [WAIT TIME: 60 seconds pair talk.] Then cold-call 4 different pairs (not volunteers first — use a class register to select): 'Kamau and Wanjiru, tell us one item from your list.' Record each response on the board in the three columns. After 4 pairs, ask the class: 'Which column has the most ideas? Which geometric property came up the most in your predictions?' Cold-call 2 more learners to respond. Do NOT evaluate predictions yet — say: 'Hold onto these predictions —	Activation of Prior Knowledge through Categorical Prediction: Learners retrieve and categorise reflection knowledge before new information is introduced. Recording predictions in three visible columns (Road/Transport, Art/Textiles, Buildings/Architecture) creates a public cognitive anchor that the class will return to at the DQB phase, allowing learners to see which predictions were confirmed, refined, or overturned — a metacognitive loop that deepens retention.	Teacher scans the room during silent writing to check that learners are producing specific examples (not just writing 'mirror'). Listen during pair talk for whether learners are naming specific properties from Lesson 2 (equal distances, perpendicular connector, congruence) — this reveals depth of recall. During cold-calling, note which learners struggle to name a property; these learners should be seated in stronger groups during the Observe phase group work.

		your group work will test them.'		
Observe Phase VIDEO: Determining reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12 Search ARES for similar readings	The teacher assigns learners to three pre-arranged groups of roughly equal size. Each group receives an Application Research Card (printed or on ARES) with: (i) a short illustrated description of their assigned real-life context; (ii) two or three guiding questions; (iii) a blank labelled-diagram template to complete. Group A (Driving Mirrors & Road Safety) studies a diagram of a matatu side mirror and a diagram of a Kenyan road intersection with a convex safety mirror on a blind corner, then sketches the object (vehicle behind), the mirror line, and the image, and annotates which property explains why the image appears the same distance behind the mirror surface. Group B (Kanga/Kitenge Design) studies a printed Kanga motif pattern, identifies the mirror lines in the design, traces the base motif and its reflection, and labels the perpendicular bisectors. Group C (Architectural Symmetry) studies photographs or line drawings of the Kenyan Parliament Buildings façade and a Mombasa Old Town mosque entrance arch, identifies vertical lines of symmetry, and sketches one architectural element with its mirror image annotated. Each group has 12 minutes to complete their diagram template and prepare a 2-minute verbal presentation. One learner acts as scribe, one as presenter, one as diagram-holder/pointer during the presentation.	Announce groups clearly and direct learners to move quickly: 'You have 30 seconds to move to your group — bring your exercise book and a pencil.' Distribute or display Application Research Cards. Say: 'Your group has exactly 12 minutes. Your job is: one, understand your real-life context; two, complete the labelled diagram on your card; three, be ready to present for two minutes. Every person in the group must be able to explain the diagram — not just the presenter.' [Set a visible 12-minute timer.] Circulate using a structured route — visit Group A first (2 min), then B (2 min), then C (2 min), then return to any group that was stuck. At each group, ask one probing question: Group A — 'Why does the driver see the vehicle behind at the SAME apparent distance as it truly is? Which property explains that?' Group B — 'Point to a mirror line in the Kanga pattern. Now point to an object point and its image — are they equidistant from that line?' Group C — 'Is the line of symmetry of the Parliament façade a mathematical mirror line? How do you know?' [WAIT TIME at each group: 15 seconds after posing the question before accepting an answer.] At the 10-minute mark, give a 2-minute warning: 'Two minutes — finish your diagram and decide who presents.' Do NOT do the thinking for any group; redirect with questions only.	Jigsaw Expert Group Investigation: Each group develops deep expertise in one application domain. During presentations in the Explain phase, all learners acquire knowledge of all three domains from their peers — a structure that mirrors (appropriately) the distributed nature of reflection across Kenyan life. The labelled diagram template enforces geometric precision: learners must identify object, mirror line, and image explicitly, preventing the common error of discussing 'symmetry' in purely visual/aesthetic terms without geometric grounding.	Teacher checks each group's diagram template during circulation: (1) Is the mirror line drawn as a dashed line, correctly positioned? (2) Are object and image equidistant from the mirror line? (3) Is at least one perpendicular connector drawn? Groups whose diagrams are missing all three features need a targeted prompt. Listen for whether group members use precise geometric language ('perpendicular bisector', 'congruent', 'equal distances') or revert to informal language ('same size', 'flipped') — this gap between informal and formal language is a key formative signal to address in the Explain phase.
Explain Phase VIDEO: Reflecting	Each group presents their application and labelled diagram to the class in a 2-minute slot (6	Introduce the presentation round: 'Each group has exactly two minutes. I will time you. After you	Peer Teaching with Cross-Context Synthesis: Learner-led presentations shift epistemic	During presentations, listen for three specific indicators of mastery: (1) Does the presenter

shapes: diagonal line of reflection <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Determine reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	minutes total for three groups). The presenter holds up or points to their diagram while explaining: (a) what the real-life context is and where in Kenya it appears; (b) which geometric properties of reflection are visible in their example; (c) how their application connects back to the anchoring phenomenon (Aisha's mirror, the matatu rear-view mirror, or the letter 'A' demonstration). After each group presents, the rest of the class has 60 seconds to ask one clarifying question or add one observation. The teacher then facilitates a 4-minute whole-class synthesis discussion, drawing connections across the three applications and returning to the prediction board from the Predict phase.	finish, the class has 60 seconds for ONE question or observation — make it mathematical, not just a comment.' [Use a visible 2-minute timer for each group.] During each presentation, stand to the side — do not interrupt. After each presentation, cold-call one non-presenting learner: 'Atieno, which of the three properties of reflection did Group A demonstrate in their diagram?' [WAIT TIME: 15 seconds.] After all three presentations, facilitate synthesis with these exact questions (cold-call, do not accept shouted answers): 1. 'What do a matatu wing mirror, a Kanga print, and the Parliament Buildings have in common — geometrically? Someone who hasn't spoken yet.' [WAIT 15 seconds, cold-call.] 2. 'Group B, your Kanga pattern had more than one mirror line. Does having multiple mirror lines change any of the three properties of reflection — or do they still all hold for each mirror line individually?' [WAIT 15 seconds.] 3. 'Connect any one of these applications back to Aisha standing in front of her bedroom mirror. What is the same? What is different?' [WAIT 15 seconds, cold-call a learner who has been quiet.] Close with: 'Every one of these applications works because of the same geometric rule — the mirror line is the perpendicular bisector of the segment joining object to image. That rule does not change whether we are on a Nairobi road, in a Mombasa mosque, or looking at a Kanga at the market.'	authority from teacher to class, increasing engagement and deepening presenter understanding. The forced connection back to the anchoring phenomenon in each presentation ensures that new application knowledge is integrated into the existing storyline rather than stored as isolated facts. The teacher's synthesis questions use the Compare-and-Contrast move to surface the invariant geometric structure (three properties of reflection) across three superficially different contexts.	name at least two of the three formal properties of reflection? (2) Does the labelled diagram correctly show perpendicular bisector relationship? (3) Can a non-presenting learner, when cold-called, identify the correct property? After the synthesis discussion, quickly survey the class with a show-of-hands question: 'Hands up if you can state the one geometric rule that is TRUE in all three applications.' Count hands — if fewer than 70% respond, re-state the rule explicitly before moving to the DQB phase. Note the names of learners who were unable to name any property during presentations for follow-up in Lesson 8.
Driving Question Board (DQB)	The teacher directs learners' attention to the class Driving Question Board, which has been	Direct learners to the DQB: 'Look at our Driving Question Board. We have been building answers since	Driving Question Board Update with Evidence Anchoring: The DQB functions as a public,	The sticky notes are the primary formative artefact of this phase. Teacher reads each note and

Creation  VIDEO: Determining reflections (advanced) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Rigid transformations: preserved properties <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	visible since Lesson 1 and carries the anchoring question: 'Why does a mirror show a perfect copy of the world — and where exactly does that copy live?' Learners individually write on a sticky note (or a scrap of paper if sticky notes are unavailable) either: (A) one question from today's lesson that they can now ANSWER on the DQB, citing evidence from group work, OR (B) one NEW question that today's applications raised for them that is not yet answered. Learners post their notes to the board. The teacher reads several aloud and facilitates a brief 3-minute class discussion to categorise them as 'Answered' or 'Still investigating.'	Lesson 1. Today you investigated three new real-world contexts. Take 90 seconds — write ONE sticky note. Either: write an answer you can now give to our driving question, and reference your group's application as evidence; OR write a new question today's work raised.' [Distribute sticky notes or scrap paper. WAIT TIME: 90 seconds of silent individual writing.] As learners post their notes, read 4–5 aloud to the class. Sort them visibly into two columns: 'We can answer this now' and 'Still investigating.' For 'We can answer this now' notes, ask the author: 'In one sentence, what is your answer and what is your evidence?' [WAIT 10 seconds before accepting.] For 'Still investigating' notes, validate: 'This is an excellent residual question — this is exactly what mathematicians and engineers ask too.' Ensure the following key answer is explicitly recorded on the DQB if no learner produces it: 'The image in every real-life mirror — matatu wing mirror, Kanga pattern, Parliament façade — appears exactly as far behind the mirror line as the object is in front, because the mirror line is always the perpendicular bisector of the object–image segment.' Draw a star next to this statement.	cumulative knowledge record for the entire unit. Requiring learners to cite specific evidence (their group's application) when posting an 'answer' note prevents superficial closure — learners must demonstrate that their answer is grounded in observed geometric relationships, not just asserted. The visible 'Still investigating' column normalises productive uncertainty and sets up Lesson 8 (the culminating lesson) as meaningful rather than merely procedural.	checks: (1) Are 'answer' notes specific and property-referenced, or vague? (2) Are 'new question' notes mathematically meaningful or merely confused? Collect all notes after the lesson and sort them — learners whose notes show persistent confusion about basic properties (e.g., 'I still don't understand why the image is the same size') need targeted support in Lesson 8. Count the ratio of answer-notes to new-question-notes: a healthy class should produce roughly 60% answer-notes and 40% new-question-notes at this stage of the sequence.
Model Building Phase  VIDEO: Reflecting shapes: diagonal line of reflection <i>Related topic: not an exact match for this lesson.</i> Source: Khan	Learners open their cumulative reflection models — the annotated diagram each learner has been building since Lesson 1, showing object, mirror line, image, and geometric annotations. Each learner adds a new layer to their model: a 'Real-World Applications Ring' — a circle or border drawn around the existing model, from which they draw three labelled	Say: 'Open your cumulative model — the diagram you have been building since Lesson 1. Today you add the final outer ring: real-world applications. Draw a large circle around your existing model. Add three arrows pointing outward — one for each application from today. Label each arrow with the application name AND one sentence explaining which	Cumulative Model Revision with Application Mapping: The 'Real-World Applications Ring' is a deliberate architectural metaphor — the geometric model at the centre is the foundation, and the applications ring is the superstructure built on it. This spatial metaphor reinforces that application does not replace or stand apart from the geometry but	Teacher circulates during model-building and checks three things for each learner's model: (1) Are all three application arrows present and correctly labelled? (2) Does each justification sentence name a specific geometric property (not just say 'reflection is used')? (3) Is the existing core model (from Lessons 2–6) still intact and correctly annotated? Collect or

Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Reflect points <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	arrows pointing outward, each arrow labelled with one of today's applications (Driving Mirrors / Road Safety, Kanga Textile Design, Architectural Symmetry) and a one-sentence geometric justification for how the application relies on the properties shown inside the model. Learners who finish early add a fourth arrow labelled 'Aisha's bedroom mirror' and annotate it with measurements like those in the Lesson 2 example (object 1.2 m from mirror, image 1.2 m behind mirror, total 2.4 m apart).	geometric property makes it work. You have 5 minutes.' [WAIT: allow 5 minutes of individual working time; circulate and observe.] After 3 minutes, cold-call one learner: 'Mwangi, read me the sentence you wrote for the driving-mirrors arrow.' [WAIT 10 seconds.] Offer corrective feedback if needed: 'Can you add the specific property — equal distances? perpendicular bisector? congruence?' After 5 minutes, ask: 'Hold up your model so your shoulder partner can see it. Partner: point to the geometric property that ALL THREE arrows rely on.' [WAIT 20 seconds for partners to examine each other's models.] Close the lesson: 'Your model now shows the full journey — from Aisha's puzzled face in her bedroom mirror, to the geometric properties that explain it, to the Cartesian rules that let us calculate it, to the real-world applications that save lives, create beauty, and shape the skyline of Kenya. In Lesson 8 we will use all of this to answer our driving question one final time — in writing.'	depends on it. Requiring a one-sentence geometric justification for each arrow forces integration rather than mere listing, and the partner-check ('point to the property that all three arrows rely on') creates a peer accountability structure that deepens encoding.	photograph 5–6 models at random after the lesson to review overnight. Models that show only decorative additions without geometric justification sentences indicate that the learner has not yet achieved the transfer of learning that this lesson targets — these learners need a focused conversation at the start of Lesson 8.
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D. TEACHER REFLECTION

1. During group presentations, were learners consistently naming specific geometric properties (equal distances, perpendicular bisector, congruence) or defaulting to informal language ('same size', 'flipped', 'symmetrical')? What does this tell me about the depth of conceptual understanding gained across Lessons 2–6, and how should I adjust the opening of Lesson 8?

2. Which of the three application contexts (driving mirrors, Kanga design, architectural symmetry) generated the richest geometric discussion, and which felt superficial or disconnected from the geometry? What would I change about the Application Research Card for that weaker group — more scaffolding, a different example, or a stronger guiding question?

3. When I read the DQB sticky notes after the lesson, what proportion showed genuine new questions versus restated confusion about basic properties from earlier lessons? Did the DQB update reveal any gaps in the foundational knowledge from Lessons 2–6 that I need to address in Lesson 8?

4. Were learners able to make the explicit connection between today's applications and the anchoring phenomenon (Aisha's mirror, the matatu rear-view mirror, the letter 'A')? If the connection felt forced or unclear, how can I re-anchor the phenomenon more concretely at the start of Lesson 8?

5. During the model-building phase, did the cumulative models show genuine integration — real-world arrows with property-referenced justification sentences — or did they show surface-level additions? For learners whose models lacked justification sentences, what targeted intervention (peer discussion, worked example, direct re-teaching) is most appropriate before the final assessment in Lesson 8?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed — through group research, sketching, and peer presentations — that driving mirrors on Kenyan matatus show an image of the vehicle behind at the same apparent distance as the real vehicle is from the mirror; that Kanga and kitenge textile patterns are constructed by reflecting a base motif across one or more mirror lines, creating bilateral symmetry; and that landmark Kenyan buildings such as the Parliament Buildings in Nairobi and mosques in Mombasa's Old Town have façades with clear vertical lines of symmetry that are geometrically equivalent to mirror lines in the plane.
What did I learn?	Every real-life application of reflection — whether a road-safety mirror, a fabric print, or a building façade — works because of the same three geometric properties established in Lesson 2: (1) the image is as far behind the mirror line as the object is in front; (2) the line joining any object point to its image is perpendicular to the mirror line; (3) the image is congruent to the object. The mirror line is always the perpendicular bisector of the object–image segment. These properties are invariant across all contexts, regardless of whether the mirror line is physical glass, a fold-line in a fabric, or an axis of architectural symmetry.
How does this explain the phenomenon?	These findings directly and completely answer the anchoring phenomenon. Aisha's reflection raises the opposite hand because reflection reverses lateral orientation across the mirror line — a geometric consequence of the perpendicular-bisector rule. Her image appears exactly as far behind the mirror as she stands in front because the mirror line is equidistant from object and image. The matatu driver's rear-view mirror shows vehicles at their true apparent distance for the same reason. The Kanga skirt pattern Aisha wears to school is itself a designed instance of reflection symmetry. Even the Parliament Buildings Aisha passes on the matatu ride display the same geometric principle. The 'invisible rule' governing where the image lands is not invisible at all — it is the perpendicular bisector of the object–image segment, and it governs every mirror in Kenya.

LESSON 8 (40 minutes): Model Building, Review and Assessment — The Mirror Fully Explained

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners synthesise all seven lessons of the reflection unit into a single cumulative annotated diagram, complete a written class activity covering all SLOs, self-assess against the KICD rubric, and articulate a complete final explanation of why the everyday plane mirror produces the image it does.
Knowledge	– State all three properties of reflection (equal distances, perpendicularity, congruence) and explain why they produce the image observed in a plane mirror. – Recall and apply the Cartesian reflection rules for mirror lines $x = 0$, $y = 0$, and $y = x$. – Explain how the mirror line is the perpendicular bisector of the object–image segment in every context — plane surface and Cartesian plane alike. – Define a line of symmetry as a special case of a mirror line that maps a figure onto itself.
Skills	– Construct a complete annotated reflection model diagram that integrates lines of symmetry, plane-surface reflection, Cartesian reflections, and reverse mirror-line construction. – Solve unseen reflection problems on both plain surfaces and the Cartesian plane within a timed class activity. – Self-assess completed work against KICD rubric descriptors and identify personal areas for further practice.
Attitudes	– Appreciate that the precise geometric properties uncovered across the unit fully and completely explain an ordinary, everyday phenomenon — the plane mirror. – Demonstrate responsibility by working independently and honestly during the written class activity.
Key Inquiry Question	How is reflection important in day-to-day activities — and can we now explain exactly why a plane mirror shows a perfect, same-sized, laterally reversed image at the same distance behind the glass as the object is in front?
Purpose in Storyline	This is the final lesson of the unit. Learners close the inquiry loop that began when Aisha noticed her reflection in her bedroom mirror and the matatu driver relied on rear-view mirrors to navigate Nairobi traffic. Every property and construction rule discovered across Lessons 1–7 is now assembled into one coherent explanation and a cumulative model diagram, and the driving question is formally answered.
Safety Notes	Learners using small plane mirrors for the final demonstration should handle glass edges carefully. Compasses have sharp points — remind learners to place them on the desk when not in use. No other specific hazards.

B. LESSON OVERVIEW

Lesson 8 is the capstone of the Sub-Strand 2.2 Reflection unit. It has three interlocking purposes: (1) model synthesis — learners produce a single large annotated diagram that brings together every concept from Lessons 1–7 (lines of symmetry, plane-surface reflection properties, Cartesian reflection rules for $x = 0$, $y = 0$, and $y = x$, and reverse mirror-line construction); (2) written assessment — learners complete a structured class activity covering all six specific learning outcomes, providing evidence for the KICD rubric; and (3) final explanation — learners explicitly and verbally answer the unit's driving question by connecting the geometry of the plane mirror to Aisha's bedroom experience and the matatu rear-view mirrors.

The lesson opens with a brief Predict phase in which learners are challenged to write, from memory, every reflection rule they have learned — a retrieval practice activity that surfaces gaps before the model-building consolidation. The Observe phase is replaced by a structured self-review: learners examine their own Lesson 1–7 notebook

work alongside a partially completed "model skeleton" printed on the ARES platform, compare it to what they recall, and annotate discrepancies. In the Explain phase learners complete the written class activity independently, then participate in a whole-class cold-call debrief that reconstructs the full explanation of the anchoring phenomenon. The DQB phase formally closes every question posted across the unit. The Model Building phase is the culminating task: learners produce their final annotated reflection model diagram and compare it to a simple sketch they made in Lesson 1, seeing in concrete visual form how far their understanding has grown.

Throughout the lesson the teacher uses the KICD assessment rubric descriptors — Exceeds Expectations, Meets Expectations, Approaches Expectations, Below Expectations — as explicit, shared success criteria. Learners self-assess and record their rubric level for each SLO indicator, developing the metacognitive self-efficacy that the KICD curriculum foregrounds. All contexts remain Kenyan: Aisha's bedroom mirror, the Nairobi matatu rear-view mirror, the bold letter "A" on the teacher's paper, and the kanga-fabric symmetry patterns introduced in earlier lessons are woven into the assessment questions and the model diagram itself.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Reflecting shapes: diagonal line of reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections (advanced) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners open their exercise books to a fresh page and, without consulting any notes, write a 'Brain Dump': every reflection rule, definition, and transformation formula they can recall from Lessons 1–7. They work silently and independently for 4 minutes, then underline any rule they are uncertain about with a wavy line. Finally, each learner counts how many distinct rules they wrote and records the number at the top of the page.	Signal the start with: 'Close every notebook except your exercise book. I want to see what is already living in your mind — no peeking at earlier work. You have 4 minutes. Go.' Set a visible timer. Circulate silently; do not answer any questions during this phase — if a learner asks, say: 'Trust what you remember — wavy-underline anything you are unsure about.' After 4 minutes call time: 'Pens down. Count your rules and write the number at the top.' Cold-call 3 learners to read their count aloud. Ask: 'Who has more than six rules? Raise your hand.' Then: 'Who has fewer than four? Keep your hand up — you are about to fix that in the next ten minutes.' WAIT TIME: 12 seconds after each cold-call before probing.	Retrieval Practice (Brain Dump): Forcing recall without notes activates long-term memory traces, identifies gaps before instruction, and primes learners to attend to the specific concepts they could not retrieve. Research shows retrieval practice produces stronger retention than re-reading notes.	Teacher scans the room during circulation, noting learners who have written fewer than three items or who have left the page nearly blank — these learners will need targeted support during the Explain phase. The cold-call count gives a rapid whole-class gauge of retention breadth.
Observe Phase  VIDEO: Determining reflections (advanced) Source: Khan Academy (English - US curriculum)  Search ARES	Learners open their notebooks to Lessons 1–7 and work through a structured 'Gap-Spotter' checklist printed on the ARES platform (or written on the board). The checklist lists seven items — one per lesson — with a small blank beside each: (1) Definition of a line of symmetry; (2) Three properties	Display the 7-item checklist on the board or ARES screen. Say: 'This checklist is your personal audit of what you know. Be honest — a circle or a cross is not a failure, it is information you can act on right now.' After learners have filled in their checklist (allow 2 minutes), say: 'Now re-read the pages that	Structured Self-Audit (Gap-Spotter Checklist): By systematically comparing recalled knowledge against a KICD-aligned checklist, learners practise metacognitive monitoring — accurately knowing what they know. This moves the Observe phase from external evidence gathering to evidence	Teacher notes how many learners have three or more crosses on the checklist — if more than a quarter of the class, spend an additional 2 minutes of whole-class review on the most commonly crossed item before proceeding to the Explain phase. Individual checklists are kept in notebooks and can be

for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12  Search ARES for similar readings	of reflection on a plane surface; (3) Construction rule — perpendicular bisector method; (4) Cartesian rule for $x = 0$; (5) Cartesian rule for $y = 0$; (6) Cartesian rule for $y = x$; (7) How to find the mirror line given object and image. For each item learners tick if they have it clearly in their notes, circle if it is partially there, or cross if it is missing. They then locate the two items they are least confident about and re-read the relevant notebook pages for 3 minutes.	cover your two weakest items. You have 3 minutes.' Circulate and whisper to learners who have crossed four or more items: 'Let me sit with you briefly during the class activity.' Reconvene with: 'Hands up — who found something in their notes that they had completely forgotten?' Cold-call 2 volunteers to share what they rediscovered. WAIT TIME: 10 seconds after each.	gathering about their own understanding, which is appropriate for a final synthesis lesson.	reviewed during marking.
Explain Phase  VIDEO: Reflecting points in the coordinate plane Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Determine reflections (advanced) Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners complete the written class activity independently (approximately 12 minutes). The activity contains six structured questions, one per SLO: Q1 — identify all lines of symmetry in a kanga-fabric hexagonal motif; Q2 — given a point $P(3, 5)$ and mirror line $y = 0$, state the image coordinates and name the property that makes the mirror line the perpendicular bisector of PP'; Q3 — draw the image of triangle KLM under reflection in the line $y = x$ on a provided Cartesian grid; Q4 — given object point $A(-2, 4)$ and image $A'(6, 4)$, construct the mirror line and write its equation; Q5 — a matatu rear-view mirror is placed so that a vehicle 8 m behind the mirror appears in the glass; how far does the image appear from the mirror and from the vehicle? Explain using a reflection property; Q6 — write two sentences explaining why Aisha's image in her bedroom mirror is the same size as Aisha. After 12 minutes, teacher leads a 5-minute cold-call debrief: 6 different learners are called to present one answer each.	Distribute or display the class activity. Say: 'This is your individual work — it shows me and you exactly where you stand against the KICD rubric. Work quietly. You have 12 minutes.' Set timer. Circulate; do not give answers but ask probing redirections quietly: 'Which Cartesian rule applies here?' or 'What does the perpendicular bisector tell you about distances?' After 12 minutes: 'Pens down. Let us hear the answers.' Cold-call Q1: 'Zawadi, how many lines of symmetry did you find, and how did you verify them?' WAIT 12 seconds. Cold-call Q2: 'Brian, what are the image coordinates and which property are you using?' WAIT 12 seconds. Continue through Q3–Q6 with different named learners. After Q6, ask the whole class: 'So — can we now fully explain why Aisha's image is the same size as her, the same distance from the mirror, but laterally reversed? Call out the three properties.' Accept choral response, then write the three properties on the board as a class-agreed summary: (1) Equal distances. (2) Perpendicularity. (3)	Evidence-to-Explanation Bridge: The written class activity serves as structured evidence that learners then collectively explain in the cold-call debrief. Connecting each answer back to the anchoring phenomenon (Aisha's mirror, the matatu driver) ensures that formal mathematical rules are always grounded in observable, meaningful context — the hallmark of sensemaking rather than rote procedure.	During circulation, teacher uses a rapid tally: mark a dot beside a learner's name on the class list for each question where they appear stuck. After the debrief, collect the written activity for marking against the KICD rubric (Exceeds / Meets / Approaches / Below Expectations for each SLO indicator). Common error patterns — e.g., swapping x and y for reflection in $y = x$, or forgetting to negate the correct coordinate for $x = 0$ vs $y = 0$ — should be noted for feedback in the next lesson or on written comments.

		Congruence (same size and shape).		
Driving Question Board (DQB) Creation  VIDEO: Reflecting shapes: diagonal line of reflection Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Reflections in a Plane Mirror (PLIX) Source: CK-12  Search ARES for similar readings	The teacher displays the class DQB that has been building since Lesson 1. Learners re-read every question posted on the board (or listed on the ARES screen). Working in pairs for 2 minutes, they decide: which questions can now be fully answered? They write a one-sentence answer on a sticky note (or in their notebook if sticky notes are unavailable) for at least two questions. The class then reconvenes and pairs share their answers aloud; the teacher physically moves answered-question cards to a 'NOW WE KNOW' column. Finally, the teacher reads the original driving question aloud — 'Why does a mirror show a perfect copy of the world — and where exactly does that copy live?' — and asks the class to answer it together in one paragraph, which the teacher scribes on the board.	Point to the DQB and say: 'Look at every question we posted across eight lessons. Some were posted by you in Lesson 1 when you barely knew the word reflection. Read them carefully.' Allow 1 minute of silent reading. Then: 'With your partner, choose at least two questions you can now answer fully. Write a one-sentence answer for each. Two minutes.' After pair work, cold-call 4 pairs: 'Fatuma and Kepha — which question did you answer and what is your answer?' WAIT 10 seconds for each pair. Move answered cards to 'NOW WE KNOW.' Then say: 'Here is the question we started with.' Read the driving question slowly. 'I am going to scribe your paragraph on the board. Give me one sentence at a time — I will cold-call.' Build the paragraph from 4–5 learner contributions. The final paragraph should include: the three reflection properties, the idea that the image lives behind the mirror at the same distance, and the reason lateral reversal occurs (the perpendicular mapping swaps left and right). Leave the paragraph on the board throughout the Model Building phase.	DQB Closure Ritual: Returning to questions posted in earlier lessons gives learners visible, concrete evidence of their own knowledge growth. Moving cards to 'NOW WE KNOW' is a public, celebratory act that reinforces the idea that science and mathematics inquiry is a progressive, cumulative process — not a single event. Co-constructing the final driving-question paragraph models how individual pieces of knowledge integrate into a coherent explanation.	Teacher listens for whether pair answers are mathematically precise (e.g., 'The image is as far behind the mirror as the object is in front because the mirror line is the perpendicular bisector of the object–image segment') or vague ('the mirror copies things'). Vague answers prompt immediate follow-up: 'Can you use the mathematical language from the board?' The co-constructed paragraph on the board serves as a class-level formative artefact.
Model Building Phase  VIDEO: Determining reflections (advanced) Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Each learner draws their Final Cumulative Reflection Model on a full page of their exercise book (or on an A4 sheet if available). The model must include, labelled and annotated: (a) a plane figure with all its lines of symmetry marked and numbered; (b) an object point P and its image P' on a plain surface, with the mirror line, the perpendicular from P to the mirror	Say: 'This is your final reflection model — the most complete diagram you will ever draw for this sub-strand. It must tell the whole story we have uncovered, from lines of symmetry all the way to finding a hidden mirror line. You have 8 minutes. Include the phenomenon box — Aisha and her mirror belong in your model.' Set timer. Circulate and offer precise,	Cumulative Model Comparison (L1 vs Final): Asking learners to physically locate their Lesson 1 diagram and compare it to their Lesson 8 model makes cognitive growth visible and tangible. This strategy — borrowed from NGSS Science and Engineering Practice 2 (Developing and Using Models) — shows learners that models are not static pictures but evolving	Teacher collects or photographs final model diagrams (or reviews them during circulation). Success indicators: all five components (a–e) are present and labelled; the three properties are explicitly stated in words, not just drawn; the Cartesian rules are written algebraically; the phenomenon box connects the geometry to Aisha's mirror. Any learner missing two or

 HTML: Lines of Symmetry (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	line, and the equal-distance markings; (c) a Cartesian grid showing reflections in all three mirror lines ($x = 0$, $y = 0$, $y = x$) for a single triangle, with the transformation rules written beside each; (d) an object–image pair where the mirror line is not given, with the perpendicular-bisector construction shown and the mirror line equation stated; (e) a small 'phenomenon box' in the corner — a sketch of Aisha standing in front of her bedroom mirror with the three properties labelled on the diagram. Learners then compare this final model to any diagram they drew in Lesson 1, write one sentence describing the most important thing that changed in their model, and self-assess their final model against the KICD rubric.	targeted praise and prompts: 'I can see your three properties labelled — now add the equal-distance tick marks.' or 'Your Cartesian grid shows $y = x$ — write the rule $(x, y) \rightarrow (y, x)$ beside it.' With 2 minutes remaining, say: 'Now find your Lesson 1 notebook page. Write one sentence: what is the single most important thing your model now shows that your Lesson 1 work could not?' Cold-call 3 learners to read their sentences aloud. WAIT 12 seconds each. Close with: 'The mirror has not changed. But your ability to explain exactly what it does — and why — has changed completely. That is what mathematics education does.' Remind learners to self-assess against the rubric in the remaining minute.	representations of understanding. The 'phenomenon box' anchors the abstract mathematics to the real-world event that started the inquiry, reinforcing the principle that every rule serves to explain something observable.	more components is flagged for a brief one-on-one follow-up. Self-assessment rubric ratings are written by the learner on the diagram itself and compared to the teacher's holistic rating after marking.
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D. TEACHER REFLECTION

1. During the Brain Dump (Predict phase), which specific rules were most frequently absent or uncertain across the class? Does this pattern suggest that a particular lesson in the unit (e.g., Lesson 5 on $y = x$, or Lesson 6 on reverse construction) needs to be restructured or given more time in future delivery?

2. When learners co-constructed the final driving-question paragraph on the DQB, did their language demonstrate conceptual understanding (referencing perpendicular bisectors, congruence, equal distances) or did it remain at a surface level ('the mirror copies the image')? What questioning moves could deepen the quality of the collaborative explanation next time?

3. How accurately did learners self-assess against the KICD rubric? Were there learners who rated themselves at 'Exceeds Expectations' but whose written activity showed gaps — or learners who rated themselves 'Below Expectations' despite strong work? What does this tell you about the metacognitive self-efficacy development in this class?

4. The phenomenon box (Aisha's mirror sketch) was intended to anchor every annotation in the real world. Did learners treat it as a genuine explanatory connection or as a decorative addition? What could be done in earlier lessons to make the phenomenon more present throughout the unit?

5. Looking at the full arc of eight lessons: which of the six specific learning outcomes (lines of symmetry; plane-surface properties; plane-surface construction; Cartesian reflection; reverse mirror-line; real-life application) showed the greatest growth, and which showed the least? How will this inform your planning for the next sub-strand (2.3 Trigonometry)?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed — through their own Brain Dump, Gap-Spotter checklist, and final model diagram — the full architecture of reflection geometry: lines of symmetry dividing plane figures into congruent halves; the three properties of plane-surface reflection (equal distances, perpendicularity, congruence); Cartesian transformation rules for $x = 0$, $y = 0$, and $y = x$; and the perpendicular-bisector method for locating a hidden mirror line. They also observed, concretely, how much their own models had grown since
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	Lesson 1.
What did I learn?	The geometry of reflection is governed by one unifying principle: the mirror line is the perpendicular bisector of the segment joining every object point to its corresponding image point. This single principle explains all three properties (equal distances, perpendicularity, congruence), generates all three Cartesian transformation rules, and makes it possible to reconstruct any mirror line from an object–image pair. Lines of symmetry are the special case where object and image are the same figure.
How does this explain the phenomenon?	The anchoring phenomenon is now fully explained. Aisha's image appears the same distance behind her bedroom mirror as she stands in front because the mirror surface is the perpendicular bisector of every object–image segment — equal distances are a mathematical consequence, not a coincidence. The image is the same size because reflection is a congruence transformation. The image is laterally reversed because the perpendicular mapping preserves distances along the mirror but flips the perpendicular direction, swapping left and right. The matatu rear-view mirror obeys identical properties. Every real-life mirror — bedroom glass, road-safety mirror, rear-view mirror — is a physical instantiation of the same perpendicular-bisector geometry that learners have now mastered.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.